Optimal Corporate Taxation and Enforcement with Market Power

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Abstract

This paper studies how market power affects optimal corporate taxation and enforcement. Higher market power typically increases the excess burden of taxation by compounding existing production distortions, pushing toward lower optimal tax rates. At the same time, the fiscal effectiveness of taxation and enforcement depends on how taxable income responds through both production and avoidance decisions, implying that neither optimal tax rates nor enforcement levels necessarily vary monotonically with market power. The relationship strongly depends on the structure of avoidance and enforcement costs. Economies of scale in avoidance or administration can make enforcement relatively more attractive for firms with high market power. The paper also highlights that firm size and market power do not always move together, implying that size-based and market-power-based policies can generate different prescriptions.

1 Introduction

The idea that large, powerful corporations do not remit their fair share of taxes has gained considerable public traction in recent decades, both aided by and reflective of news reports of many of these top firms remitting little to no federal taxes. For example, in 2020 it was prominently reported that dozens of Fortune 500 companies remitted no federal income taxes the past year, with a subset of those also remitting nothing in the previous two years.[1] A 2023 Government Accountability Office (GAO) study reported that on average about a quarter of profitable large corporations remitted zero taxes between 2014 and 2018, and average effective tax rates hovered around 15% until dropping to single digits due to the Tax Cuts and Jobs Act of 2017 (TCJA).[2]

What constitutes a fair share is inherently subjective and further complicated by the distinction between legal avoidance and illegal evasion. Moreover, many of these large firms remitting little to no taxes in a given year are reporting losses or carrying forward or back losses.[3] Despite this, perceptions of under-taxation remain strong. In a yearly Gallup poll taken between 2004 and 2019, the fraction of surveyed Americans who responded that corporations "pay too little" in taxes consistently stood at two-thirds, while the fraction believing corporations provide their "fair share" remained at one fifth.[4]

This paper examines efficiency based justifications for targeting tax or enforcement policy on market power. It develops a model of optimal business taxation and enforcement with imperfectly competitive industries. An industry with market power will already feature suboptimal production even in the absence of taxation. This raises excess burden and tends to reduce optimal tax rates as market power increases. However, economies of scale in either avoidance technology or administrative burden can still make it advantageous to increase relative enforcement on these firms or industries. This paper also outlines a distinction between market power and firm size. While often strongly correlated, the two characteristics do not always overlap and can carry different implications for optimal policy.

The influence of the excess burden factor on optimal policy diminishes with increased cost deductibility. Under a pure profit tax in which all costs are fully deductible, production behavior is undistorted by the level of effective taxation, and tax policy imposes no excess burden regardless of market structure. While most corporate income taxes are profit taxes, there are reasons to believe that effective fractions of cost deductibility fall below 100 percent. For example, no OECD country has a corporate tax system that features full loss offset, and many have limited depreciation allowances on capital expenditures.[5] Thus, this excess burden factor likely remains relevant to some extent under real world profit taxes.

[1] https://www.forbes.com/sites/tommybeer/2021/04/02/more-than-50-major-us-corporations-including-nike-and-fedex-paid-no-federal-taxes-last-year/?sh=3a7b338521d3

[2] https://www.gao.gov/products/gao-23-105384

[3] While carryback was disallowed by the TCJA 2017, the Coronavirus Aid, Relief, and Economic Security (CARES) Act in 2020 allowed firms to carryback losses 5 years for tax years beginning in 2018 to 2020.

[4] https://news.gallup.com/poll/1714/taxes.aspx

[5] The model in this paper does not directly incorporate loss years as it is static and firms must be weakly profitable to stay in business. But such ideas can be connected if the model is taken as a long-run perspective on profits.

The key fiscal margin is the elasticity of reported income, which decomposes into a true income (or tax base) elasticity and an avoidance elasticity. Unless there is a pure profit tax, the true income elasticity reflects a combination of market structure, the definition of the tax base, and underlying demand and cost conditions. As a result, its relationship with market power is generally ambiguous and can vary with whether changes in market power operate through concentration or conduct.

More structurally stable is the avoidance elasticity. This elasticity is governed by the interaction between the scale of the tax base and the curvature of avoidance costs. When avoidance costs are sufficiently concave in the tax base, larger tax bases are associated with stronger avoidance responses to both taxation and enforcement. In many standard environments, increases in market power are also associated with larger tax bases, so market power and avoidance responses tend to move together. However, this link is not robust across all settings. Differences in deductibility, demand, and conduct-based changes in market power can decouple market power, tax base, and firm size, leading to potentially opposing policy implications.

Lower avoidance cost-base convexity increases the responsiveness of avoidance to both taxation and enforcement, but it affects the two instruments asymmetrically. When avoidance is highly responsive to scale, taxation induces stronger behavioral responses in the form of increased avoidance, making it relatively more distortionary. Enforcement, however, works to reduce avoidance rates, broadening the reported tax base and providing a fiscal benefit. This shifts the optimal policy mix away from statutory taxation and toward greater reliance on enforcement as the tax base becomes larger or more responsive through the avoidance channel.

In contrast, when avoidance costs are more convex in the tax base, scale plays a weaker role in shaping avoidance behavior in the sense that large tax bases no longer imply a disproportionate increase in avoidance responsiveness. Instead, the avoidance response becomes relatively stronger for smaller tax bases. As a result, the enforcement advantage shifts towards firms with smaller tax bases. Only in the knife-edge case where avoidance costs are linear in the tax base does the (avoidance-related) tradeoff between taxation and enforcement become independent of scale or market power.

Economies of scale on the administrative side reinforce these forces. If enforcement costs are concave in the tax base, then the marginal cost of enforcement falls with firm tax bases, making it more efficient to concentrate enforcement on large tax bases rather than spread enforcement across many small ones. This provides a distinct channel through which enforcement may be disproportionately directed toward high market power firms (provided they also have high tax bases) even in the absence of larger avoidance behavior.

The implications of these results are important for policy-relevant discussion. Attitudes toward firm remittance behavior have not been limited to popular discourse. Auditing rates have consistently been correlated with firm size, with the largest firms facing audit probabilities several times higher than those of the smallest firms. As illustrated in Table 1, while overall enforcement intensity has declined over time, the relative gradient in auditing probability by firm size has remained broadly stable.

This pattern is consistent with the model's implications under plausible conditions, but it is not a direct test of the underlying mechanism. In particular, firm size is only an adequate proxy for the relevant tax base when size, market power, and taxable income move closely together. When they diverge, for example under conduct-based variation in market power, targeting based on size may not align with where avoidance incentives or fiscal distortions are strongest.

However, the same empirical pattern can still be optimal even without a close correspondence between size and the tax base if enforcement costs depend primarily on measures such as assets rather than taxable income itself. In that case, higher audit probabilities for large firms would still reflect cost efficiency.

Table 1: Auditing Rates by Firm Size (%)

Firm Size (Balance Sheet Assets)	2011	2013	2015	2017
\$1–\$999,999	1.00	0.81	0.55	0.32
\$1,000,000–\$99,999,999	2.48	2.03	1.94	1.45
\$100,000,000–\$999,999,999	20.50	17.31	16.15	9.41
$>$ \$1,000,000,000	48.13	37.95	32.57	24.11

Data Source: IRS Data Book 2021, Table 17. Categories used by IRS redefined into the above four broader groups. More recent years are still potentially subject to change due to the 3-year rolling auditing process and so are not included.

More recently, there have been measures with more explicit targeting of top firms remitting more taxes. Two of the provisions of the landmark Inflation Reduction Act of 2022 in the US were to increase funding for the IRS by \$80 billion over the next decade, with \$45 billion earmarked for increasing and improving enforcement activities, and the introduction of a 15 percent Corporate Alternative Minimum Tax (CAMT) on corporations averaging over one billion dollars in income over a three year span. Among the plans for improvement was a stronger focus on tax system equity through enforcement on high income individuals and large firms who appear to remit considerably less taxes than expected.[6] This increase in funding was not a unanimous decision, illustrated by an attempt by House Republicans to block the funding in January 2023, and later by a \$20 billion reduction in the funding as a part of the Fiscal Responsibility Act of 2023. These conflicting decisions within a short time span reflect the lack of consensus on how enforcement should be structured and how it relates to the broader tax system.

The model in this paper does not directly feature an audit probability as illustrated above and as in the prototypical Allingham and Sandmo (1972) setup. Instead, a reduced form enforcement parameter enters into the firm's avoidance costs, similar to Kopczuk et al. (2016) and Keen and Slemrod (2017). The government can either spend resources on enforcement to increase these costs for the firm or adjust the statutory tax rates. A firm in an industry with a fixed number of

[6]See introductory quote, from U.S. Secretary of the Treasury Janet Yellen's letter to the IRS: https://home.treasury.gov/system/files/136/JLY-letter-to-Commissioner-Rettig-Signed.pdf

competitors then chooses output levels and avoidance rates to maximize profits subject to these policy tools.

The analysis begins with a tax system in which both tools are fully differentiable by industry, in the spirit of the Ramsey (1927) commodity tax model, which allows the underlying forces to emerge clearly. Policymakers, however, often face instrument constraints, and most corporate tax rates are set uniformly. This paper therefore also analyzes a uniform tax system that still permits differentiated enforcement. While targeting is imperfect under this constraint, the core analysis carries through.

This paper connects to several strands of the literature. It first extends the optimal taxation literature that integrates agent evasion decisions and tax agency administrative decisions (Kaplow, 1990; Cremer and Gahvari, 1993; Dharmapala et al., 2011; Keen and Slemrod, 2017). A particularly relevant result from Kaplow (1990) mirrors the preceding discussion: despite similarities between the two policy tools, differences in behavioral distortions and resource costs create a tradeoff such that both instruments can be used optimally.

These studies connect to those that introduce market power into optimal tax problems (Besley, 1990; Delipalla and Keen, 1992; Stern, 1987; Chetty, 2009; Kaplow, 2021; Eckhout et al., 2021), a literature that highlights the relevance of firm entry and exit, tax rate pass-through, profit retention, and pre-existing output distortions. Pass-through in particular has received recent attention in relation to both market power (Anderson et al., 2001; Weyl and Fabinger, 2013; Pless and van Benthem, 2019; Miklós-Thal and Shaffer, 2021; Adachi and Fabinger, 2022; Ritz, 2024) and evasion (Kopczuk et al., 2016).

Finally, this paper relates to the broad literature on the role of firms in the tax system (Kopczuk and Slemrod, 2006; de Paula and Scheinkman, 2010; Slemrod and Velayudhan, 2018). Much of the optimal tax literature largely abstracts away from the direct role of firms. In reality, however, firms remit the vast majority of taxes, and modeling their avoidance behavior and the enforcement responses it invites is integral to understanding the tax system.

The empirical evidence on the connection between market power and avoidance rates remains limited but has attracted growing attention in the accounting literature. Kubick et al. (2015) find a negative association between product market power and effective tax rates, their proxy for tax avoidance, and hypothesize that persistent profitability due to market power allows firms to predict future income streams more accurately, raising the value of avoidance strategies. Martin et al. (2022) examine the reverse direction and find that firms engaging in tax avoidance achieve higher sales, contributing to higher concentration ratios. While these studies do not conclusively establish the true relationship between market power and avoidance, they provide suggestive evidence.

The rest of this paper is structured as follows. Section 2 details the model for each type of agent: consumer, firm, and government. Section 3 breaks down the welfare and fiscal effects of taxation and enforcement and how these change with market power. Section 4 discusses optimal policy, both in levels and in the tradeoff between the two tools. Section 5 restricts the fully differentiated tax rates to a uniform system and examines how the intuition from the Ramsey setup carries over.

Section 6 illustrates features of the model and examines these interactions through a simulated environment. Section 7 discusses the main results and concludes. The majority of mathematical derivations appear in the Appendix.

2 Model Setup

There is a set of K industries with potentially varying degrees of competition in the economy. Each industry produces a unique good k.[7] Consumers face a tax-inclusive price q_k for each good, while producers receive the net-of-tax price $p_k = q_k(1 - \tau_k)$.

2.1 Consumer Problem

A representative consumer chooses a consumption bundle $\vec{x}$ and leisure, or equivalently labor supply L. Labor serves as the numeraire good, and the wage is normalized to 1. The consumer also receives all profits Π_j from each industry. The consumer solves

$$\begin{aligned} &\max_{\vec{x},L}\ u(\vec{x}, L) \\ &\text{s.t. } I + L + \sum_j \Pi_j \geq \vec{q} \cdot \vec{x}. \end{aligned} \tag{1}$$

The term I denotes nonlabor income and defines the marginal utility of income, $\frac{\partial v}{\partial I} = \alpha$, where v is indirect utility. The consumer takes prices and profits as given and does not internalize any effect of individual decisions on either variable. To simplify the analysis and isolate the key mechanisms, the model assumes quasilinear utility. This eliminates income effects in product demand and rules out cross-price effects across markets.

2.2 Firm Problem

A firm in industry k chooses output y_k and an avoidance rate γ_k to maximize profits, subject to a tax rate τ_k and an enforcement rate β_k. The term "avoidance" is used throughout to encompass both legal avoidance and illegal evasion, and γ_k captures both margins. This simplification imposes an important limitation, since firms may face different marginal costs for legal avoidance and illegal evasion in practice. As a result, a single margin abstracts from potentially important distinctions in both firm tax planning and enforcement policy.

Firms receive the producer price p_k and may deduct a fraction $\mu \in [0, 1]$ of production costs $\phi_k(y_k)$ from taxable income. Define the pre-avoidance tax base as $b_k \equiv y_k q_k - \mu\phi_k(y_k)$. Avoidance costs $H_k(b_k, \gamma_k, \beta_k)$ depend on the firm's tax base, the avoidance rate, and enforcement. Total costs equal production and avoidance costs, $C_k(y_k, b_k, \gamma_k, \beta_k) = \phi_k(y_k) + H_k(b_k, \gamma_k, \beta_k)$. Subsequent notation suppresses the industry index in the cost function when taking derivatives. For example, $H_\gamma \equiv \frac{\partial H_k(b_k,\gamma_k,\beta_k)}{\partial \gamma_k}$. However, the underlying cost functions may still differ across industries.

[7]Alternatively, industries are partitioned into isolated markets and k represents a good-market combination.

In order to obtain feasible interior solutions for output and avoidance, the model assumes that ϕ and H are continuous and twice differentiable in all arguments, with $\phi_y > 0$ for $y \geq 0$, $H_\gamma > 0$ and $H_{\gamma\gamma} > 0$ for $\gamma > 0$. Since higher enforcement should increase the cost of avoidance, the model also imposes $H_\beta > 0$ and $H_{\gamma\beta} > 0$ for $\gamma > 0$.[8] The firm solves

$$\max_{y_k,\gamma_k} y_k q_k(x_k) - C_k\big(y_k, b_k(y_k), \gamma_k, \beta_k\big) - \tau_k(1-\gamma_k)b_k(y_k). \tag{2}$$

If $\mu = 0$, no costs are deductible and firms face a pure turnover or revenue tax. If $\mu = 1$, all production costs are deductible and firms face a pure profit tax. The parameter μ therefore determines the relevant tax base: firms shift reported revenues under a turnover tax and reported profits under a profit tax. This tax base enters the cost function with the implicit assumption that avoidance costs may differ depending on whether firms shift revenues or profits. The specification excludes avoidance costs from these deductible expenses. In practice, many avoidance-related expenditures may be deductible, but including this deductibility has little effect on the results while substantially complicating the expressions.

Each market contains a fixed number N_k of identical competitors and a conduct parameter $\theta_k \in [0,1]$ that jointly determine market power. Following Ritz (2024), $\theta_k^c \equiv \frac{\theta_k}{N_k}$ represents a composite index of market power, which ranges from $\theta_k^c = 1$ under monopoly to $\theta_k^c = 0$ under perfect competition. Firm behavior satisfies a standard first order condition equating marginal revenue and effective marginal cost,

$$q_k\left[1 - \frac{\theta_k^c}{\varepsilon_{x_k}}\right] = \frac{1-\mu\omega_k}{1-\omega_k}\phi_y, \tag{3}$$

where $\varepsilon_{x_k} \equiv -\frac{\partial x_k}{\partial q_k}\frac{q_k}{x_k}$ denotes the (positively defined) elasticity of demand and $\omega_k = (1-\gamma_k)\tau_k + H_r$ denotes the operative tax rate faced by the firm.[9,10] A useful benchmark arises under a pure profit tax, in which case the tax system does not distort production decisions. Output is then independent of both the statutory tax rate and the avoidance margin, and the firm chooses the same quantity as in a no-tax setting.

For nonzero marginal costs, the first-order condition imposes the restriction

$$1 - \frac{\theta_k^c}{\varepsilon_{x_k}} > 0. \tag{5}$$

[8] An additional mild regularity condition is that H_γ is log-concave, which includes all power functions with powers greater than 1. This rules out excessively rapidly accelerating convexity in the cost function.

[9] The operative tax rate is the equivalent tax rate in a no avoidance world that leads to the same production behavior. This differs slightly from the effective tax rate collected by the government, which excludes the avoidance cost term H_r.

[10] Rearranging the production condition shows the elasticity-adjusted Lerner index equals the composite measure of market power:

$$\frac{q_k - \frac{1-\mu\omega_k}{1-\omega_k}\phi_y}{q_k}\varepsilon_{x_k} = \theta_k^c. \tag{4}$$

This condition requires firms to operate only where demand is sufficiently elastic, namely $\varepsilon_{x_k} > \theta_k^c$. Under monopoly, this reduces to the standard requirement that production occurs on the elastic portion of demand ($\varepsilon_{x_k} > 1$). More generally, the allowable region expands as either the number of firms increases or conduct becomes more competitive, since both reduce θ_k^c.

The profit maximizing avoidance condition is

$$\tau_k b_k = H_\gamma(b_k, \gamma_k, \beta_k) \tag{6}$$

which equates the marginal benefit and marginal cost of avoidance. The marginal benefit of avoidance, $\tau_k b_k$, increases linearly in the firm's tax base. The properties of the optimal avoidance rate therefore depend on how marginal costs scale with the tax base.

If avoidance costs are linear in firm's tax base, such that $H(b_k, \gamma_k, \beta_k) = b_k H(1, \gamma_k, \beta_k)$, then the avoidance rate does not depend on the tax base or market structure as marginal benefits and costs scale equivalently. If instead $H(\cdot)$ is convex in the tax base, marginal avoidance costs increase more than proportionally with scale, implying a lower optimal avoidance rate for firms with larger tax bases. If $H(\cdot)$ is concave, the opposite holds. To characterize these cases more precisely, define the elasticity of marginal avoidance costs with respect to the firm's (true) tax base as

$$\varepsilon_{H_\gamma}^b \equiv \frac{H_{\gamma b} b}{H_\gamma}. \tag{7}$$

An elasticity equal to 1 corresponds to constant returns to scale (CRS) in the avoidance technology. Values above 1 correspond to decreasing returns to scale (DRS), while values below 1 correspond to increasing returns to scale (IRS). These three cases are referred to throughout as CRS, DRS, and IRS avoidance technologies, respectively. Empirical evidence on this elasticity remains limited, although tax authorities may have information relevant for identifying its likely range.[11] The following Lemma translates these types of technologies into relationships between market power and optimal avoidance rates.

Lemma 1. *Consider the relationship between the avoidance rate γ_k and market power, summarized by $\theta_k^c \equiv \theta_k / N_k$. An increase in θ_k^c may arise either from a decrease in N_k or an increase in θ_k.*

(i) When θ_k^c increases through a reduction in N_k (holding θ_k fixed), the relationship between γ_k and market power decreases with $\varepsilon_{H_\gamma}^b$: it is positive for $\varepsilon_{H_\gamma}^b < 1$, zero at $\varepsilon_{H_\gamma}^b = 1$, and negative for $\varepsilon_{H_\gamma}^b > 1$.

(ii) Suppose instead that θ_k^c increases through an increase in θ_k, holding N_k fixed. Under a pure profit tax ($\mu = 1$), the result in part (i) holds exactly. For $\mu < 1$, the sign of the relationship between γ_k and θ_k^c coincides with part (i) if and only if

$$1 - \theta_k^c > \frac{1-\mu}{1-\mu\omega_k}(\varepsilon_{x_k} - \theta_k^c). \tag{8}$$

[11] Bachas et al. (2019) find that larger firms exhibit higher compliance, though jointly with enforcement intensity. The avoidance technology here captures ex ante avoidance capacity, for which evidence remains limited.

Otherwise, the relationship reverses.

The intuition for part (i) is that changes in N_k primarily operate through the scale of firms' tax bases. As the number of firms declines, the per-firm tax base increases, strengthening the role of scale in the avoidance technology. The response of avoidance therefore depends on how marginal avoidance costs scale with the tax base, summarized by $\varepsilon^b_{H_\gamma}$. When marginal costs scale proportionally with the tax base (CRS), the tax base has no additional effect on the optimal avoidance rate. When marginal costs are convex (DRS), firms with larger tax bases face disproportionately higher marginal avoidance costs, leading to lower avoidance, and vice versa under concavity (IRS).

Part (ii) introduces an additional channel that operates when market power increases through changes in conduct rather than changes in the number of firms. In this case, N_k is fixed, so variation in market power operates through changes in markups and equilibrium outcomes within the market. Higher conduct raises prices but reduces output. The per-firm tax base may therefore rise or fall depending on which effect dominates and the relevant tax base. This implies that greater market power does not systematically translate into a larger tax base. Under a pure profit tax, the relevant tax base is profits. Profits do unambiguously increase with softer conduct. In this case, relationship between conduct and avoidance mirror that of concentration and avoidance.

Lastly, the interactions between output distortion, market power, and the type of avoidance technology can create numerous dependencies that complicate the key ideas at work. As such, this paper simplifies results by only considering CRS avoidance when discussing revenue taxation. Profit taxation results, which can ignore the production side, will feature statements on the types of avoidance technologies.

2.3 Government

The government chooses industry-specific tax rates τ_j and enforcement efforts β_j to maximize the utility of the representative agent subject to the budget constraint

$$\sum_j \tau_j(1-\gamma_j)B_j - N_j A(b_j, \beta_j) \geq G, \tag{9}$$

where $B_j = N_j b_j$ is the pre-avoidance taxable income (tax base) in industry j, $A(\cdot)$ represents per-firm administrative costs, and G is an exogenous revenue requirement that determines the government budget multiplier λ.[12] Defining total reported taxable income as $Z_j \equiv (1-\gamma_j)B_j$, the government's Lagrangian is then

$$\mathcal{L} = v(q) + \lambda \left[\sum_j \tau_j Z_j - N_j A(b_j, \beta_j) - G \right]. \tag{10}$$

[12]In within-market comparisons, such as changes in the number of firms within an industry, aggregate equilibrium outcomes also change. In this sense, λ is endogenous to market structure. Throughout, these comparisons assume either that the rest of the economy is sufficiently large so that such changes have a negligible effect on λ, or equivalently that one is comparing otherwise identical economies that differ only in market power.

A practical complication is that administrative costs likely contain a fixed component at the firm level. Explicitly incorporating fixed enforcement costs, however, would introduce an extensive-margin enforcement decision in which the government optimally excludes firms below a threshold level of taxable income (Dharmapala et al., 2011). The analysis abstracts from this margin in order to preserve a continuous intensive-margin enforcement problem and smooth comparative statics. Instead, the model captures similar economic forces through cost function curvature. When $A(\cdot)$ is concave in or independent of the per-firm tax base b_j, average enforcement costs decline with scale, generating a fixed-cost-like force on the intensive margin in which enforcement becomes relatively more cost-effective for firms with a larger base.

The fully differentiated system above is useful for deriving intuition. However, most real world corporate tax systems use a common statutory tax rate across all firms and industries. Section 5 therefore considers an alternative specification in which the government constrains tax rates to be uniform across industries while continuing to allow industry-specific enforcement policies. This reflects the reality that enforcement authorities typically have greater discretion in allocating enforcement resources than in setting statutory tax rates, which are more heavily shaped by legislative constraints.

2.4 Between Market vs. Within Market Comparisons

As in Ritz (2024), the analysis distinguishes between two types of market power comparisons. The between markets approach takes two markets with the same observed elasticities but differences in market power. The within markets approach takes a market (or two markets with the same structural parameters) and changes the market power. In this latter approach, elasticities themselves may also change as market power changes. This paper considers both types of comparisons as they may have different implications. The prior is useful for cross-sectional comparisons between markets with similar observables (excepting for market power), while the latter is useful for thinking about singular markets that may be changing concentration or conduct over time.

3 Market Responses to Taxation and Enforcement

Important to the discussion of the optimal tax and enforcement rates id how firms and therefore the markets respond to these tools. As in standard optimal tax settings, the problem involves balancing marginal welfare effects against marginal fiscal effects.

3.1 Welfare Effect

As the representative consumer owns all firms, the welfare effect equals the sum of the direct price effect and the profit change. By the envelope theorem, re-optimization of the consumption bundle does not affect welfare at the margin, and income and cross-price effects vanish by assumption.

Thus, the welfare effect is

$$\frac{1}{\alpha}\frac{dv}{dT_k} = -\frac{\partial q_k}{\partial T_k}x_k + \frac{\partial \Pi_k}{\partial T_k} \tag{11}$$

where v is the indirect utility of the consumer and $T_k \in \{\tau_k, \beta_k\}$ represents one of the tax tools.

Define the pass-through semi-elasticity of taxation as $\rho_k \equiv \frac{\partial q_k}{\partial \tau_k}\frac{1}{q_k}$. Then the aggregate profit response for an industry k is

$$\frac{\partial \Pi_k}{\partial \tau_k} = \underbrace{-Z_k}_{Mechanical} + \underbrace{(1-\omega_k)(1-\theta_k^c)\rho_k x_k q_k}_{Competition} \tag{12}$$

The first term captures the mechanical effect of taxation on profits, equal to the negative of aggregate reported income. The second term captures the behavioral response through prices and output adjustments due to competitors. Under monopoly ($\theta_k^c = 1$), the competitive term vanishes and only the mechanical effect remains, consistent with the envelope condition at the firm level. With multiple firms ($N_k > 1$), individual firms exert some influence over prices, generating a nonzero behavioral component. As competition intensifies, this term increases in magnitude and approaches the mechanical effect in the limit of perfect competition.[13]

Similarly, the impact of a change in the enforcement rate on aggregate profits is

$$\frac{\partial \Pi_k}{\partial \beta_k} = -N_k H_\beta + (1-\omega_k)(1-\theta_k^c)\rho_k^\beta x_k q_k \tag{13}$$

where the same deconstruction into a mechanical and competitive adjustment occurs. The mechanical term captures the direct increase in expected compliance costs induced by higher enforcement, while the second term reflects adjustments in prices and quantities that arise through strategic interaction in the market.

Using the profit response expression, the welfare effect in response to a tax change can be written as

$$\frac{1}{\alpha}\frac{dv}{d\tau_k} = -Z_k - \rho_k x_k q_k\left[\omega_k + (1-\omega_k)\theta_k^c\right] \tag{14}$$

Breaking down this expression, the first term is the reported income in the industry, which exactly equals the mechanical gain in tax revenue. The second term captures the additional welfare effect beyond the mechanical transfer from firms and consumers to the government, representing the marginal excess burden of taxation. Dividing through by reported income converts this expression to the welfare effect per mechanical dollar raised:

$$\frac{1}{\alpha Z_k}\frac{dv}{d\tau_k} = -1 - EB_k \tag{15}$$

[13] This statement applies starting from a zero-tax baseline. At nonzero tax rates, even highly competitive markets may exhibit nontrivial profit responses due to tax-base interactions.

where the marginal excess burden per dollar of revenue is given by

$$EB_k = \rho_k \frac{x_k q_k}{Z_k} \left[\omega_k + (1-\omega_k)\theta_k^c\right] \tag{16}$$

Similarly, the change in welfare due an increase in enforcement per dollar of reported income is

$$\frac{1}{\alpha Z_k} \frac{dv}{d\beta_k} = -\frac{N_k H_\beta}{Z_k} - EB_k^\beta \tag{17}$$

where EB_k^β is equivalent to EB_k except in replacing the tax pass-through with the enforcement pass-through. While this expression does not have as natural of a breakdown into an expected transfer of a dollar from the consumers or firms to the government and the excess cost, similar logic can be applied. The first term is the impact on profits per dollar of mechanical tax revenue and is thus the "standard" burden. The second term is the welfare loss above and beyond this mechanical impact.

The presence of both market power and avoidance frictions complicates empirical measurement of these objects. Pass-through rates are commonly estimated, but identification of θ_k^c requires markup estimation, for which a recent literature has developed methods using production data (De Loecker and Warzynski, 2012; De Loecker et al., 2020). More challenging is the estimation of the operative tax rate ω_k, which depends on both avoidance behavior and the structure of the avoidance cost function. Finally, the marginal compliance cost H_β is typically unobserved, although recent work has begun to measure administrative and regulatory compliance costs directly (Harju, 2019; Trebbi et al., 2022).

Despite these challenges, the framework delivers sharp comparative statics under additional structure. Under CRS avoidance, if the two industries have the same observed elasticity of demand, supply, and curvature, then the excess burden is higher under higher market power. If the two industries have the same structural demand cost and supply parameters, then excess burden is still higher under higher market power. This represents the simple intuition that market power introduces its own distortions, and additional effective taxation compounds that distortion either through direct taxation or through enforcement.

To formalize these comparisons, define demand elasticity, demand curvature, and marginal cost elasticity as $\varepsilon_k \equiv -\frac{dx_k}{dq_k}\frac{q_k}{x_k}$, $\xi_k \equiv -x_k' \frac{q_k''(x_k)}{q_k'(x_k)}$, and $\eta_k \equiv y_k \frac{\phi_k'(y_k)}{\phi_y(y_k)}$. The following lemma summarizes the resulting comparative statics.

Lemma 2. *Under a revenue tax and CRS avoidance,*

(i) For two industries with the same observed elasticities $(\varepsilon_x, \xi_x, \eta_s)$, the excess burden of revenue taxation is higher in the industry with more market power if and only if

$$\left(1 - \frac{\theta_k^c}{\varepsilon_k}\right) \Delta_k > \frac{\omega_k + (1-\omega_k)\theta_k^c}{1-\omega_k} \left(1 - \xi_{x_k} + \frac{1}{\varepsilon_k}\right) \tag{18}$$

where $\Delta_k = 1 + \theta_k^c(1-\xi_k) + \eta_k(\varepsilon_k - \theta_k^c)$. The same is true for the (price-related) burden of *enforcement.*

(ii For a given market, the marginal excess burden increases as market power increases if and only if

$$(1-\omega_k)\rho_k + \frac{d\rho_k}{d\theta_k^c}[\omega_k + (1-\omega_k)\theta_k^c] > 0 \tag{19}$$

The enforcement expression is similar but with the enforcement pass-through.

(iii) Compliance costs are decreasing/constant/increasing with market power under IRS/CRS/-DRS avoidance technology so long as market power increases the tax base. (This is always true under a profit tax.)

The second part reflects a within-market comparison in which market power affects elasticities, curvature, and marginal costs endogenously. As a result, the sign of the effect depends more heavily on the joint structure of demand and supply and on whether market power arises through concentration or conduct. However, some clear benchmarks follow directly. At zero taxation and enforcement, imperfect competition generates higher excess burden than perfect competition. Under linear demand and constant marginal cost, a monopolist exhibits a higher marginal excess burden whenever $\omega_k < \frac{1}{2}\left(1 - \frac{1}{\varepsilon_k}\right)$.

Finally, compliance costs are decreasing in market power if and only if avoidance costs are concave in the tax base (IRS avoidance technology).[14] Intuitively, although higher market power increases avoidance in levels, a sufficiently large tax base can even the compliance costs out per dollar as long as avoidance costs are not overly convex.

3.2 Fiscal Effect

Define the elasticities of reported taxable income with respect to the tax rates and enforcement rates, respectively, as

$$\varepsilon_{Z_k} \equiv \frac{\partial Z_k}{\partial(1-\tau_k)}\frac{1-\tau_k}{Z_k}, \qquad \varepsilon_{Z_k}^{\beta} \equiv \frac{\partial Z_k}{\partial \beta_k}\frac{\beta_k}{Z_k} \tag{20}$$

Let $\varepsilon_{B_k} \equiv \frac{\partial B_k}{\partial(1-\tau_k)}\frac{1-\tau_k}{B_k}$ and $\varepsilon_{B_k}^{\beta} \equiv -\frac{\partial B_k}{\partial \beta_k}\frac{\beta_k}{B_k}$ be the corresponding elasticities of the true tax base. These elasticities mirror the elasticities of taxable income and total earned income in individual income models with avoidance and evasion such as Chetty (2009). The change in tax revenue collected from an industry $R_k \equiv \tau_k Z_k - N_k A(b_k, \beta_k)$ due to a change in the tax rate is

$$\frac{dR_k}{d\tau_k} = Z_k\left(1 - \frac{\tau_k}{1-\tau_k}\left[\varepsilon_{Z_k} - \frac{A_b B_k}{\tau_k Z_k}\varepsilon_{B_k}\right]\right) \tag{21}$$

The 1 (multiplied by Z_k) represents the mechanical revenue raised. The bracketed term in both expressions provides a net tax revenue elasticity. The term subtracted off the standard income

[14] In this statement and in Lemma 2, the $(1-\gamma_k)$ term in the tax base is omitted, so the expression refers to compliance cost per unit of true tax base. This term cancels in the elasticity-based formulation used later and does not affect comparative statics.

elasticity represents an endogenous cost-saving effect. As effective taxation increases, the potential tax base may shrink and scaled down resources needed for enforcement. This provides some degree of cost-saving and dampens the magnitude of the total effect. Similarly, the change in tax revenue due to a change in the enforcement is

$$\frac{dR_k}{d\beta_k} = Z_k \frac{\tau_k}{\beta_k} \left[\varepsilon^{\beta}_{Z_k} + \frac{A_b B_k}{\tau_k Z_k} \varepsilon^{\beta}_{B_k} - \frac{N_k A_\beta}{Z_k} \right] \tag{22}$$

Contrary to the tax elasticities, larger enforcement elasticities indicate a positive for the government. A higher value of $\varepsilon^{\beta}_{Z_k}$ would imply that enforcement is especially effective in reducing avoidance without also substantially reducing the true tax base due to higher effective taxation. If the true tax base does shrink, the second term provides automatic cost saving as in the previous expression.

Under a true profit tax, ε_{B_k} and $\varepsilon^{\beta}_{B_k}$ equal zero as quantity decisions are undistorted. The ratio $\frac{A_b B_k}{\tau_k Z_k}$ is an approximation of enforcement costs per dollar of mechanical tax collected. If administrative costs are largely fixed relative to tax base, then this ratio becomes negligible and again the marginal resource costs are unchanged.

The last term is the marginal administrative costs of enforcement. Enforcement is most beneficial when the total marginal costs $N_k A_\beta$ are small compared to taxable income. For within market comparisons, it will be seen that the object that really matters is the marginal enforcement costs to the true base b_k. Enforcing firms with large true bases is cost-effective when there are economies of scale in enforcement, i.e., when $A(\cdot)$ is concave in the tax base argument.

3.2.1 Reported Taxable Income

The elasticities of reported income can be decomposed into the avoidance response and the true tax base response:

$$\varepsilon_{Z_k} = \varepsilon_{\gamma_k} + \varepsilon_{B_k}, \qquad \varepsilon^{\beta}_{Z_k} = \varepsilon^{\beta}_{\gamma_k} + \varepsilon^{\beta}_{B_k} \tag{23}$$

where $\varepsilon_{\gamma_k} = \frac{\partial(1-\gamma_k)}{\partial(1-\tau_k)} \frac{1-\tau_k}{1-\gamma_k}$ and $\varepsilon_{\gamma_k} = \frac{\partial(1-\gamma_k)}{\partial \beta_k} \frac{\beta_k}{1-\gamma_k}$ The avoidance response can be further decomposed into a direct and another tax base response, but this will remain as one elasticity for now. The magnitude and direction of the avoidance elasticity determines the absolute effectiveness of enforcement, the relative effectiveness of enforcement versus taxation, and how enforcement should change with market power.

The government increasing the tax rate increases the marginal benefit of avoidance by the firm's current potential tax base b_k. As the firm increases its avoidance, it raises its marginal cost by $\frac{\partial \gamma_k}{\partial \tau_k} H_{\gamma\gamma}$. The direct change in the avoidance rate is thus $\frac{b_k}{H_{\gamma\gamma}}$. As H is a function of the tax base itself, it becomes clear that the firm most benefits when H is concave in b such that the initial net benefit is higher for firms with higher tax base.

A second channel arises because changes in the tax rate also affect the operative tax wedge faced by the firm. This alters production decisions and therefore generates an endogenous change in the underlying tax base. The previous discussion isolates the direct channel holding the tax base fixed. The following lemma characterizes only this direct component.

Lemma 3. *Under CRS avoidance technology, there is no relationship between market power and the direct avoidance response to either taxes or enforcement. Under IRS (DRS) avoidance technology, the direct response of avoidance to taxation is increasing (decreasing) in market power, while the direct response to enforcement is decreasing (increasing) in market power. Under a pure profit tax ($\mu = 1$), this direct effect coincides with the total avoidance response.*

Under profit taxation, the true taxable income response is zeroed out since the production decision is unaffected by the effective tax rate. This implies that the reported income response is equivalent to just the avoidance response. Thus, the relationship between the reported income response and market power exactly follows the relationship between avoidance response and market power, as described in Lemma 3.

Under a revenue tax ($\mu = 0$), the true tax base response does not drop out. The true taxable income response simplifies to

$$\varepsilon_{B_k} = (1 - \tau_k)\rho_k(\varepsilon_k - 1) \tag{24}$$

and a similar expression for the related enforcement elasticity. Note that the taxable income here is just reported revenue. For inelastic demand, $\varepsilon_k < 1$, an increase in the tax (enforcement) rate decreases (increase) the reported income elasticity. This is because an increase in the effective tax rate increases industry revenue under inelastic demand. This broadening of the tax base is a beneficial fiscal effect and thus pushes the elasticity in the respective beneficial directions for each tool.

More important is how these expressions relate to market power. Consider the constant avoidance technology case. The avoidance response is independent of the firm's tax base. Then the only thing that matters is how the pass-through rate and the demand elasticity change with market power, i.e., how the elasticity of true revenue changes with market power. The following proposition characterizes the relationship with market power in both a between markets and within market setting.

Proposition 1. *Under a profit tax, the elasticity of taxable income and the elasticity of enforcement are independent/increases/decreases with market power under CRS/IRS/DRS avoidance technology. Under a revenue tax with CRS technology,*

(i) If $(\varepsilon_x, \xi_x, \eta_s)$ are controlled for, the elasticity of taxable income is increasing with market power if and only if $\varepsilon_{x_k} > 1$.

(ii) For the same market increasing in market power, the elasticity of taxable income is increasing with market power if and only if

$$\frac{d\varepsilon_{x_k}}{d\theta_k^c}\rho_k + (\varepsilon_{x_k} - 1)\frac{d\rho_k}{d\theta_k^c} > 0 \tag{25}$$

Under a profit tax, avoidance responses drive reported income responses. Under CRS avoidance, there is no differentiation of firm avoidance behavior by market power and so the elasticity of income is independent of market power. For IRS avoidance, more profitable firms having a comparative advantage in avoidance leads to a larger response margin.

Under a revenue tax, there is ambiguity in the direction of the true tax base response as firm revenue is not monotonic with respect to market power. Thus, depending on the elasticity of demand (and demand curvature), it can be possible for market power to have either a positive or negative impact on the real income response.

3.2.2 Administrative Costs

Recalling Table 1 and papers such as Almunia and Lopez-Rodriguez (2018) and Bachas et al. (2019), real world tax authorities often disproportionately target large firms. One simple justification for this policy prescription is economies of scale in enforcement costs, i.e., that

$$N_k A_\beta(\beta_k, b_k) > A_\beta(\beta_k, B_k) \tag{26}$$

A potential point of contention is whether administrative costs are a function of the tax base or a function of firm size as determined by total assets or sales. Under profit taxation, it can be the case that the tax base (profits) and are not one-to-one. This, however, would point more in favor of market-power based taxation. Conditioning on size, the firm with more market power is likely to have a larger potential tax base, and the "bang for your buck" logic applies even stronger.

Under revenue taxation, the tax base and firm size are aligned, but firm size and market power may not be. Appendix A and the discussion following Lemma 1 demonstrate that softer conduct may lead to higher market power but a *lower* per-firm tax base under elastic demand. In this sense, the administrative cost argument would push in favor of larger firms with less market power.

4 Optimal Rates and Tradeoff

Having separately derived the marginal welfare and fiscal effects, conditions for optimal policy follow directly. The optimal tax and enforcement equate their two respective marginal effects.

4.1 Optimal Tax Rates and Enforcement Rates

The optimal tax and enforcement rate thus satisfy the following expressions:

Proposition 2. *Given an enforcement rate, the optimal tax rate in an industry k is*

$$\frac{\tau_k}{1-\tau_k} = \frac{1 - \frac{\alpha}{\lambda}\left[1 + EB_k\right]}{\varepsilon_{Z_k} - \frac{A_b B_k}{\tau_k Z_k}\varepsilon_{B_k}} \tag{27}$$

Given a tax rate, the optimal enforcement rate in an industry k is

$$\beta_k = \frac{\tau_k z_k \left(\varepsilon^{\beta}_{Z_k} - \frac{A_b B_k}{\tau_k Z_k}\varepsilon^{\beta}_{B_k}\right)}{A_\beta + \frac{\alpha}{\lambda}\left(H_\beta + z_k E B_k^{\beta_k}\right)} \tag{28}$$

Both expressions follow standard Ramsey logic.[15] Optimal policy trades off mechanical revenue gains against behavioral distortions and enforcement costs. In the tax expression, higher fiscal elasticity in the denominator increases the marginal cost of raising revenue and therefore reduces the optimal tax rate. In the enforcement expression, higher responsiveness of reported income in the numerator increases the marginal benefit of enforcement and therefore raises the optimal enforcement level.

The structure of these expressions clarifies how market power and avoidance jointly affect policy. As shown in Proposition 1, the true tax base response depends on demand and supply fundamentals and can therefore be ambiguous in sign. By contrast, the avoidance response depends more directly on the avoidance technology (Lemma 3). This channel therefore affect optimal tax and enforcement in opposite ways: stronger avoidance responsiveness tends to reduce the optimal tax rate but increase the attractiveness of enforcement.

The excess burden terms in both expressions act uniformly to reduce optimal policy levels. Higher taxation or enforcement increases the effective wedge faced by firms, amplifying production distortions. Under mild conditions, this excess burden is increasing in market power, reflecting the fact that markets with market power already operate away from the competitive benchmark. Additional fiscal wedges therefore compound existing inefficiencies.

Under a pure profit tax, these relationships simplify substantially. Since production remains undistorted, both the competition-related excess burden and the true tax base elasticities disappear. The relationship between the conditional optimal tax rate and market power is then determined entirely by how avoidance behavior varies with market power, as characterized in Lemma 3.

The case of enforcement is more nuanced due to administrative costs. When enforcement costs exhibit economies of scale in the tax base (for example due to fixed costs or concavity in $A(\cdot)$), enforcement becomes more attractive in high-market power industries. This result can hold even under DRS avoidance technology. By contrast, if administrative costs are sufficiently convex in the tax base, diseconomies of scale can dominate, potentially making enforcement less attractive in high–market power industries even when avoidance responds strongly. Since avoidance responses are themselves endogenous to enforcement intensity, these forces also feed back into the optimal tax rate at the joint optimum.

Corollary 2.1. *Under a profit tax, the conditional optimal tax and enforcement rates are given by*

$$\frac{\tau_k}{1-\tau_k} = \frac{1-\frac{\alpha}{\lambda}}{\varepsilon_{Z_k}}, \qquad \beta_k = \frac{\tau_k z_k \varepsilon^{\beta}_{Z_k}}{A_\beta + \frac{\alpha}{\lambda} H_\beta} \tag{29}$$

[15] In a perfectly competitive economy without avoidance or enforcement, expression (27) reduces to the standard Ramsey rule.

where $\varepsilon_Z = \varepsilon_{\gamma_k}$ and similarly for enforcement. Under constant scale avoidance technology, the optimal profit tax and enforcement rate are independent of market power. Under increasing (decreasing) returns to scale avoidance technology, the tax (enforcement) is negatively (positively) related to market power all else equal.

This corollary implies that the elasticity of reported taxable income with respect to the tax rate is a sufficient statistic for the optimal profit tax rate, independent of the underlying avoidance technology. By contrast, the enforcement elasticity alone is not sufficient to determine the optimal enforcement rate, since enforcement also depends on marginal administrative costs and compliance technology through A_β and H_β.

4.2 Policy Choice: Taxation vs. Enforcement

The previous section characterized the optimal level of taxation and enforcement conditional on the other instrument. This section instead examines how the planner optimally trades off between the two policy tools. At the joint optimum, it must be true that a revenue neutral adjustment between taxation and enforcement has no first order welfare impact.

Proposition 3. *The tradeoff between taxation and enforcement satisfies*

$$\frac{\varepsilon^{\beta}_{Z_k} - \frac{A_b B_k}{\tau_k Z_k}\varepsilon^{\beta}_{B_k} - \frac{\beta_k}{\tau_k z_k}A_\beta}{1 - \frac{\tau_k}{1-\tau_k}\left(\varepsilon_{Z_k} - \frac{A_b B_k}{\tau_k Z_k}\varepsilon_{B_k}\right)} = \frac{\beta_k}{\tau_k}\frac{\frac{H_\beta}{z_k} + EB^{\beta}_k}{1 + EB_k} \tag{30}$$

This result is directly comparable to the tradeoff discussed in Keen and Slemrod (2017).[16] The left-hand side captures the relative fiscal effectiveness of enforcement compared to taxation, while the right-hand side captures their relative welfare costs.

A larger enforcement elasticity or a larger tax elasticity both shift policy toward enforcement. The former increases the fiscal return to enforcement, while the latter raises the distortionary cost of taxation. Relative to Keen and Slemrod (2017), the key new feature is the presence of the two excess burden terms. These terms imply that the optimal policy mix depends not only on avoidance responses, but also on how each instrument distorts production through imperfect competition. As shown earlier, the distinction between the two excess burden terms is driven primarily by differences in pass-through rates.

The inequality form of Proposition 3 also provides a simple characterization of welfare-improving reforms away from the optimum. If the left-hand side exceeds the right-hand side, then taxation generates a larger marginal welfare loss per dollar of revenue than enforcement. In that case, welfare increases through a revenue-neutral shift toward greater enforcement and lower statutory taxation.

Economies of scale in either compliance or administrative costs further strengthen the case for enforcement because the marginal welfare cost of enforcement rises more slowly with tax base size.

[16] This proposition is a modified version of Proposition 4 in their paper.

Conversely, sufficiently strong diseconomies of scale weaken the case for enforcement and increase the relative attractiveness of statutory taxation.

Under a pure profit tax, production distortions disappear and the excess burden terms drop out. The optimal policy mix is then determined entirely by how avoidance responses vary with market power. In particular, stronger scale economies in avoidance imply that the government should rely relatively more on enforcement as market power rises.

Under CRS avoidance technology, equation (30) simplifies to

$$\frac{\varepsilon^{\beta}_{Z_k} - \frac{A_b B_k}{\tau_k Z_k}\varepsilon^{\beta}_{B_k} - \frac{\beta_k}{\tau_k z_k} A_\beta}{1 - \frac{\tau_k}{1-\tau_k}\left(\varepsilon_{Z_k} - \frac{A_b B_k}{\tau_k Z_k}\varepsilon_{B_k}\right)} = \frac{\beta_k}{\tau_k}\frac{H_\beta(1, \gamma_k, \beta_k)}{1-\gamma_k} \tag{31}$$

In this case, welfare distortions no longer determine how the optimal policy mix varies with market power. Instead, the relationship depends entirely on how the relative fiscal effectiveness of taxation and enforcement changes with market power. The preceding two findings are summarized in the following corollary.

Corollary 3.1. *Under a profit tax and tax-base-neutral administrative costs, the ratio of taxation to enforcement is constant, decreasing, or increasing under CRS, IRS, and DRS avoidance technology, respectively. Greater concavity in administrative costs further lowers this ratio as market power rises. Under a revenue tax with CRS avoidance technology, the optimal ratio coincides with the revenue-maximizing ratio.*

5 Uniform Taxation

Most corporate tax systems impose a largely uniform statutory tax rate.[17] This section considers a uniform tax system while allowing enforcement to vary across industries. Much of the same intuition still applies, but enforcement now only partially internalizes the fiscal externalities of the tax. First, the optimal uniform tax rate satisfies

$$\frac{\tau}{1-\tau} = \frac{\sum_j Z_j \left[1 - \frac{\alpha}{\lambda}\left[1 + EB_j\right]\right]}{\sum_j Z_j \left[\varepsilon_{Z_j} - \frac{A_b B_j}{\tau Z_j}\varepsilon_{B_j}\right]} \tag{32}$$

This expression shows that the optimal uniform tax rate depends on how fiscal weights co-vary with behavioral responses across industries. Industries with high reported income receive greater weight in both the welfare and fiscal terms. When high-income industries also exhibit large behavioral elasticities or large excess burdens, they contribute disproportionately to the distortion from taxation, which lowers the optimal uniform rate. More generally, a positive correlation between the tax base and either the tax elasticity or excess burden reduces the optimal uniform tax rate relative to a setting with independent heterogeneity. Thus, in the observed cross-section, if high

[17] Firms may still face heterogeneous effective tax burdens due to deductions, credits, industry-specific provisions, and related tax rules.

market power firms are strongly correlated with high reported incomes, then previous discussions would typically expect a lower uniform rate compared to the independence case.

Under a pure profit tax the, relationship simplifies to

$$\frac{\tau_k}{1-\tau_k} = \frac{1-\frac{\alpha}{\lambda}}{\varepsilon_Z} \tag{33}$$

where $\varepsilon_Z = \frac{\sum_j Z_j \varepsilon_{Z_j}}{\sum_j Z_j}$ is the weighted average reported income elasticity. As in the differentiated case, while reported income remains the observed object, avoidance responses entirely determine the magnitude of this elasticity. Similar to the previous expression, the uniform rate is lower when high market power industries are the largest by reported income and use increasing returns to scale avoidance technology.

Because the tax rate is uniform, all cross-industry targeting must occur through enforcement. To characterize optimal enforcement differences across industries, divide the enforcement conditions for two industries m and n:

$$\frac{\beta_m}{\beta_n} \frac{A_\beta(\beta_m, b_m) + \frac{\alpha}{\lambda}[H_\beta(\beta_m, b_m, \gamma_m) + EB_m^\beta]}{A_\beta(\beta_n, b_n) + \frac{\alpha}{\lambda}[H_\beta(\beta_n, b_n, \gamma_n) + EB_n^\beta]} = \frac{z_m}{z_n} \frac{\left(\varepsilon_{Z_m}^\beta - \frac{A_b b_m}{\tau z_m} \varepsilon_{B_m}^\beta\right)}{\left(\varepsilon_{Z_n}^\beta - \frac{A_b b_n}{\tau z_n} \varepsilon_{B_n}^\beta\right)} \tag{34}$$

This condition closely mirrors the fully differentiated case. The key difference is mechanical. Because the tax rate is constrained to be uniform, the ratio of tax rates no longer appears. In the fully differentiated system, cross-industry targeting occurred directly through the ratio $\frac{\tau_m}{\tau_n}$.

Under uniformity, the common tax rate τ lies between the differentiated tax rates. Because τ and β are complements in the cost structure, a higher tax rate increases the marginal return to enforcement, causing the optimal enforcement rate to rise as well. For industries in which the differentiated tax rate was relatively low, the uniform tax rate represents an increase, which raises the marginal return to enforcement and therefore increases β in that industry. Conversely, for industries in which the differentiated tax rate was relatively high, the uniform rate represents a decrease, reducing the marginal return to enforcement and therefore lowering β.

Enforcement therefore moves in the same direction as the tax rate under the transition to uniformity: industries whose tax rates rise receive more enforcement, while industries whose tax rates fall receive less. This pattern does not arise because enforcement "picks up the slack" from the loss of tax differentiation, which would treat τ and β as substitutes and imply the opposite adjustment. Instead, it reflects the complementarity between the two instruments. Enforcement shadows the tax rate, so if the uniformity constraint pushes τ upward relative to its differentiated benchmark, optimal enforcement should rise as well.

Under a profit tax, the excess burden terms drop out, leaving

$$\frac{\beta_m}{\beta_n} = \frac{A_\beta(\beta_n, b_n) + \frac{\alpha}{\lambda} H_\beta(z_n, \gamma_n, \beta_n)}{A_\beta(\beta_m, b_m) + \frac{\alpha}{\lambda} H_\beta(z_m, \gamma_m, \beta_m)} \frac{z_m \varepsilon_{Z_m}^\beta}{z_n \varepsilon_{Z_n}^\beta} \tag{35}$$

Connecting back to the relationship between tax base size and avoidance yields the following implication:

Proposition 4. *Under a uniform profit tax and IRS/CRS/DRS avoidance technology, the enforcement rate is relatively higher/constant/lower with more market power than it would be under a differentiated profit tax.*

The joint choice of taxation and enforcement can equivalently be viewed as the choice of operative tax rates ω_k and the policy ratio $\frac{\tau_k}{\beta_k}$. With fully differentiated instruments, the government can independently select the desired operative tax rate in each industry and implement it using the most efficient combination of taxation and enforcement. Uniform taxation removes this flexibility. If the government targets operative tax rates, it must adjust enforcement to compensate, which generally leads to inefficient instrument ratios. Conversely, if it maintains efficient instrument ratios, operative tax rates will no longer align with their desired levels.

This discussion also implies that the operative tax rate selected for each industry need not change between the differentiated and uniform systems. Since firms make production decisions based on operative tax rates, market allocations and production equilibria remain unchanged across the two systems. The welfare loss from uniform taxation instead arises entirely through the avoidance and enforcement margins.

With $|K| > 2$, the optimal uniform tax rate need not lie between the differentiated tax rates of any two industries. For example, it is possible that $\tau > \tau_m > \tau_n$. Proposition 4 therefore concerns the relative distance between enforcement rates rather than their absolute levels. Both enforcement rates may increase or decrease in absolute terms even as the relative gap between them widens or narrows.

6 Simulations

To provide more intuition for how the cost structure shapes the results, this section illustrates the model using a concrete functional form. Consider the specification

$$C(y, b, \gamma, \beta) = \phi(y) + \beta b^{\rho} \gamma^2$$

Administrative costs take the form $A(\beta, b) = \beta^2 b^{\delta}$ where δ governs economies of scale in enforcement. Lower values of δ imply stronger economies of scale in administration, while $\delta = 1$ implies that enforcement costs scale proportionally with the tax base. For simplicity, $\phi(y)$ implies constant marginal production costs.

The simulations use concentration, measured by the number of firms, as the source of variation in market power. Appendix C presents analogous simulations based on differences in market conduct. Each simulation compares two industries with identical demand and cost structures that differ only in the degree of market power. This difference alone generates distinct equilibrium outcomes in prices, output, revenue, and profits, even under identical tax parameters. The exercise therefore isolates the within-market comparisons discussed earlier.

Each figure set compares the optimal policy under a fully differentiated tax system and a uniform tax system, while always allowing enforcement to vary across industries. Solid lines denote the differentiated regime, and dashed lines denote the uniform regime. Panels (a) through (d) report optimal statutory tax rates, enforcement rates, tax-to-enforcement ratios, and firm size measured by firm revenue.

6.1 Profit Taxation

The first two simulation sets in Figures 1 and 2 fix conduct at $\theta = 1$ and vary the number of firms.[18] At each point, industry 2 contains one additional firm and is therefore more competitive. These figures both set $\delta = 1$, so administrative cost economies of scale are temporarily ignored. Under CRS avoidance technology (Figure 1), avoidance does not vary with market power. If administrative costs also scale proportionally with the tax base, the government has no reason to differentiate either tax rates or enforcement rates across industries. As a result, the policy ratio τ/β remains identical across industries. Because the differentiated optimum already equalizes policy across industries, imposing a uniform tax constraint generates no additional distortion. The uniform and differentiated systems therefore coincide. As discussed in Corollary 2.1, if administrative costs did not scale proportionately to the tax base, then there would be an incentive to increase enforcement on higher market power firms.

Figure 2 instead sets $\rho = 0.8$, which generates IRS avoidance technology. In the differentiated system, the more competitive industry receives both a higher optimal tax rate and a higher optimal enforcement rate. More concentrated firms avoid more income under IRS technology, which shrinks the reported tax base and expands the pool of recoverable avoided income. This raises both the tax elasticity and the enforcement elasticity.

The increase in the tax elasticity dominates, causing the concentrated industry to face a lower optimal effective tax burden overall. At the same time, the lower tax-to-enforcement ratio shows that the government still relies relatively more on enforcement than taxation within that industry. In other words, the planner prefers to implement a given effective tax burden through enforcement rather than statutory taxation for firms with greater market power.

Under uniform taxation, the government can no longer differentiate statutory tax rates directly. Enforcement must therefore adjust to maintain the same operative tax rate. As a result, the concentrated industry receives a higher enforcement rate than the competitive industry even though it faced lower enforcement under the differentiated regime. The results reverse under DRS avoidance technology. This also illustrates the statement in Proposition 4: under uniform taxation, the enforcement ratio between industries moves closer together when $\frac{\tau_1}{\tau_2} < 1$, and moves further apart when $\frac{\tau_1}{\tau_2} > 1$, relative to the differentiated benchmark.

Across both figure sets, the policy implications of market power align closely with those of firm

[18] By doing so, these simulations represent standard Cournot competition.

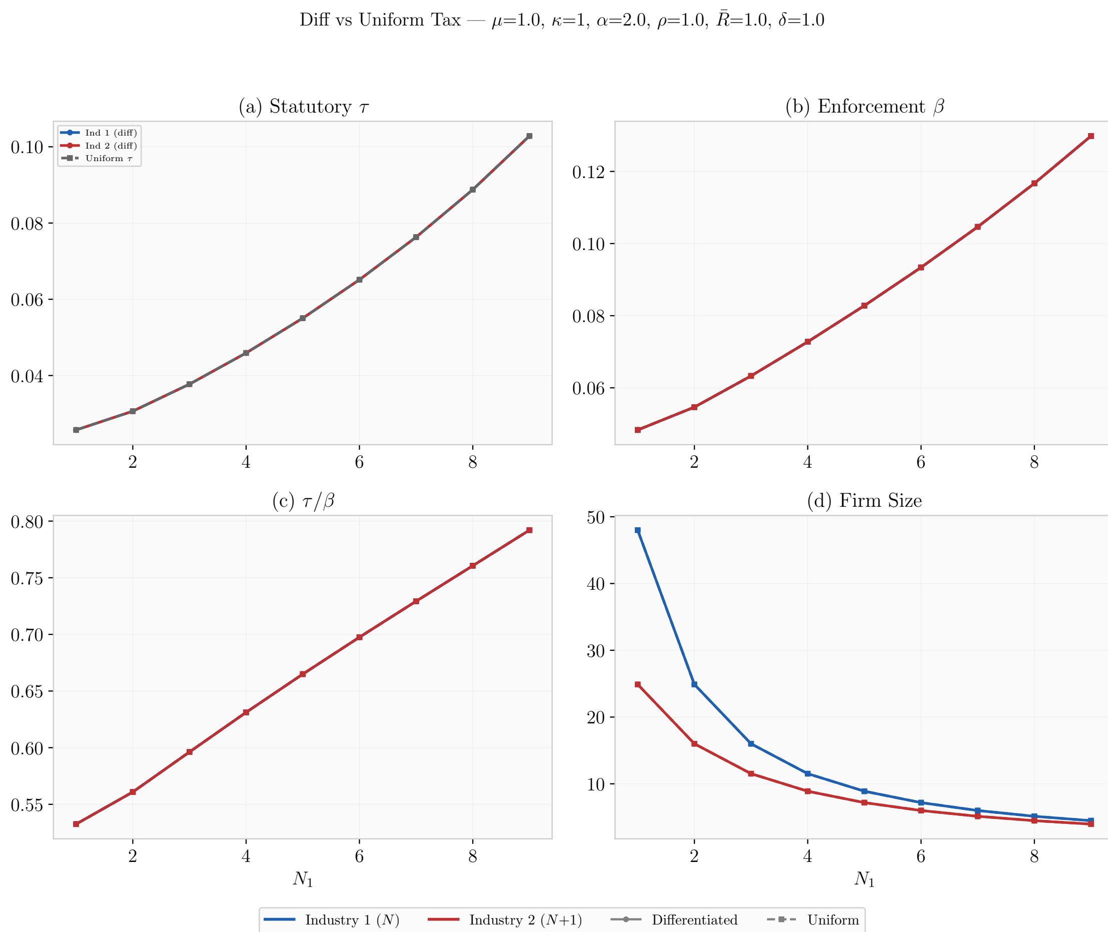

Figure 1: Optimal Policy by Number of Firms under Profit Taxation and CRS Avoidance

Notes: At each parameter value (x-axis), the two points represent the same structural industry but with one industry having an additional firm. Firm size is defined as the average revenue per firm. This simulation features a profit tax with CRS avoidance technology and proportional administrative costs.

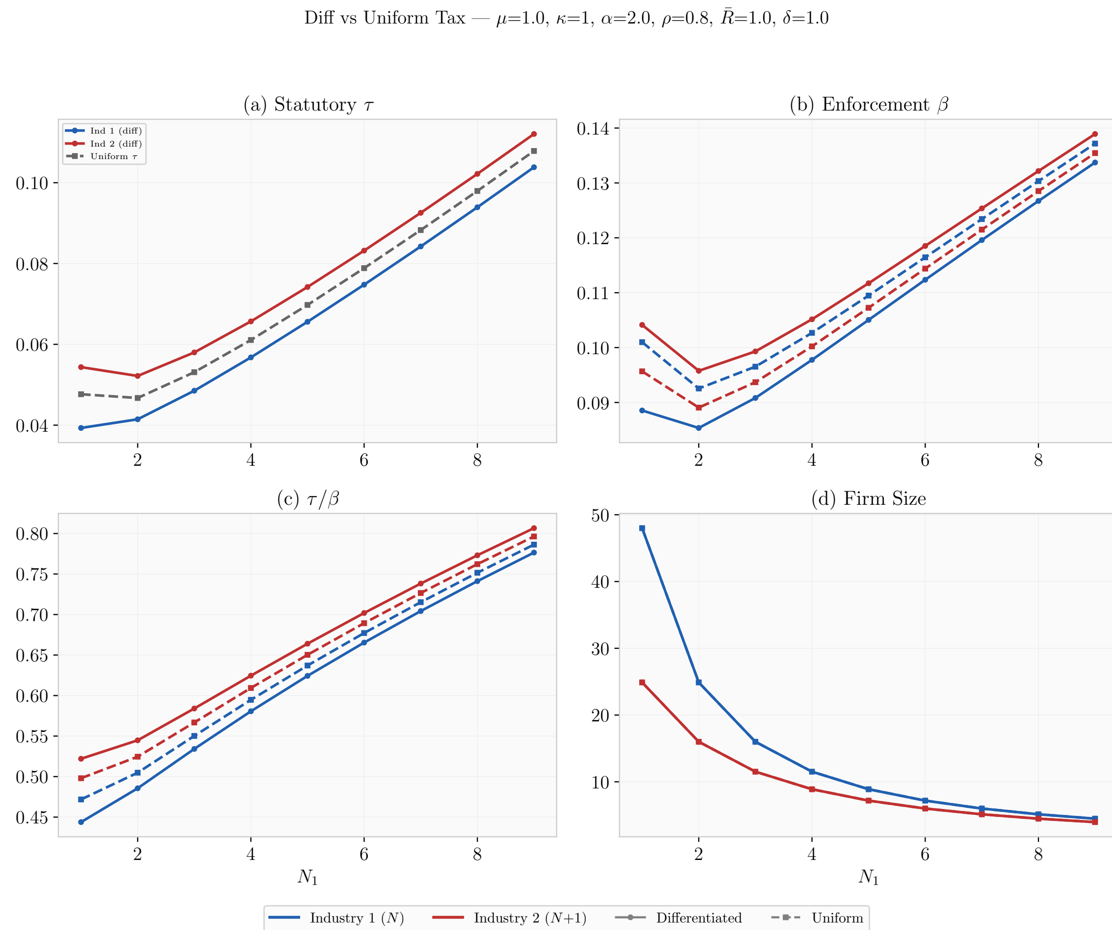

Figure 2: Optimal Policy by Number of Firms under Profit Taxation and IRS Avoidance

Notes: At each parameter value (x-axis), the two points represent the same structural industry but with one industry having an additional firm. Firm size is defined as the average revenue per firm. This simulation features a profit tax with IRS avoidance technology and proportional administrative costs.

size because concentration shifts the tax base in the same direction. In the CRS case, neither market power nor firm size affects avoidance behavior or fiscal responses, so there is no basis for differentiation in either taxation or enforcement. In the IRS case, higher market power and larger firms both generate stronger avoidance responses and a larger pool of recoverable income. As a result, enforcement targets more concentrated industries and larger firms under a uniform tax system. With either IRS avoidance technology or economies of scale in administrative costs, market-power-based prescriptions generate enforcement patterns similar to those observed in Table 1.

6.2 Revenue Taxation

The simulations in Figures 3 and 4 consider the other polar case of revenue taxation, which helps isolate how excess burden considerations shape optimal policy. Even under CRS avoidance technology (Figure 3), the planner still differentiates policy across industries. The reason is that both taxation and enforcement generate welfare costs, and these costs vary systematically with market power.

In particular, excess burden applies to both instruments. Higher market power increases distortions in the more concentrated industry, which lowers the optimal levels of both taxation and enforcement relative to more competitive industries. Moving to a uniform tax system partially reallocates this burden across industries, effectively reducing the relative excess burden in the more competitive sector. This shift changes the optimal tax-to-enforcement mix and alters how the planner trades off the two instruments across industries.

Figure 4 introduces (rather extensive) concavity in administrative costs by setting $\delta = 0$. This adjustment strengthens the role of enforcement for firms with greater market power, consistent with earlier results on economies of scale in administration. In some regions of the comparative static, this effect is strong enough to reverse the ordering of optimal tax rates across industries, again reflecting the complementarity between taxation and enforcement.

This figure also highlights that policy prescriptions based on market power are not necessarily monotonic in concentration. The relationship between market power and optimal taxation or enforcement changes across regions of the parameter space, with transition points around $N = 5$ for taxation and $N = 2$ for enforcement where the ranking across industries switches.

Figure 8 in the Appendix in a conduct comparative static demonstrates a separate mechanism. In that case, the mapping between market power and firm size itself becomes non-monotonic, so that increases in market power can correspond to either higher or lower firm size depending on the region of the comparative static. Despite this, enforcement preserves the rank ordering in market power over the entire parameter space. This makes clear that market power and enforcement maintain a one-to-one ordering in this example, whereas the relationship between market power and firm size does not.

Taken together, these results separate two sources of policy ambiguity. Figure 4 shows that optimal taxation and enforcement need not vary monotonically with market power even when firm

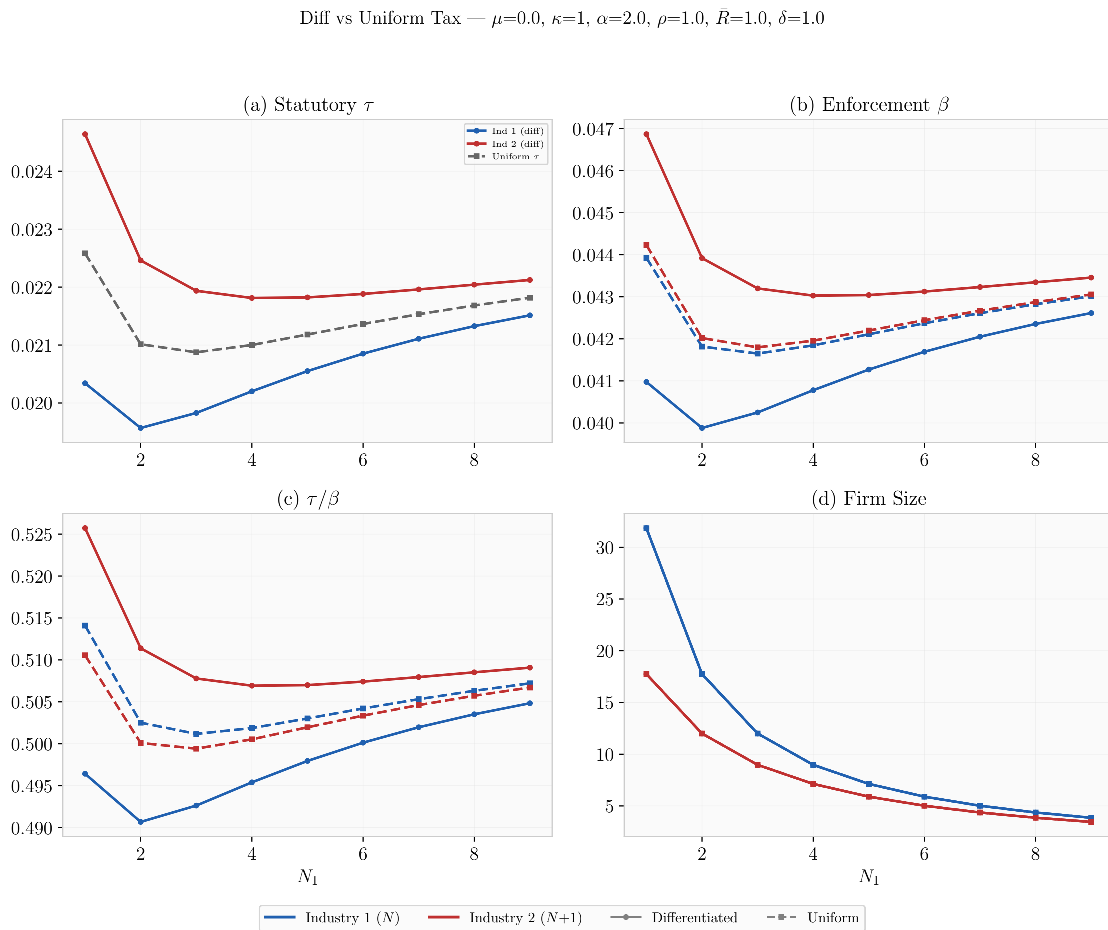

Figure 3: Optimal Policy by Number of Firms under Revenue Taxation and CRS Avoidance

Notes: At each parameter value (x-axis), the two points represent the same structural industry but with one industry having an additional firm. Firm size is defined as the average revenue per firm. This simulation features a revenue tax with CRS avoidance technology and proportional administrative costs.

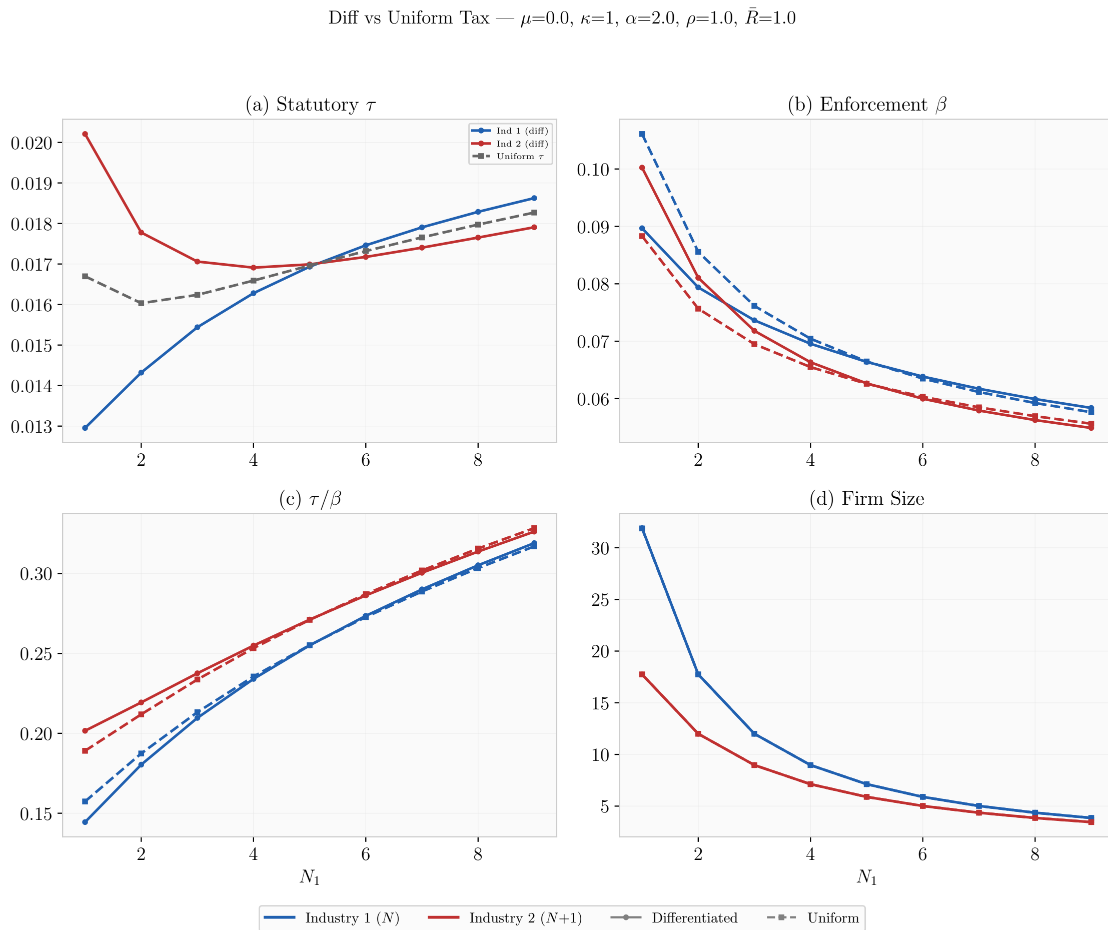

Figure 4: Optimal Policy under Revenue Taxation and CRS Avoidance with Concave Admin Costs

Notes: At each parameter value (x-axis), the two points represent the same structural industry but with one industry having an additional firm. Firm size is defined as the average revenue per firm. This simulation features a profit tax with CRS avoidance technology and concave administrative costs.

size and market power move one-to-one. The appendix figure goes further by showing that once the mapping between market power and firm size itself breaks down, policy prescriptions based on either object can diverge even more sharply.

7 Discussion and Conclusion

Whether market concentration or supernormal firm size should trigger greater enforcement intensity or higher statutory tax rates has become an increasingly important policy question. Real world enforcement patterns often align with firm size, as larger firms tend to receive disproportionately more audits. This paper shows that such patterns cannot be interpreted as a general optimal policy rule, though may be rationalized under very plausible assumptions.

The analysis highlights two primary forces that shape optimal taxation and enforcement. First, effective taxation generates excess burden that typically rises with market power, since taxes compound preexisting production distortions. This force pushes toward lower optimal tax rates in industries with greater market power. Second, the elasticity of reported taxable income depends on the underlying avoidance technology as well as the demand and cost environment. Because these elasticities determine the fiscal effectiveness of taxation and enforcement, the overall relationship between market power and optimal policy is generally ambiguous.

Despite this general ambiguity, certain structural features push toward stronger enforcement of larger or more concentrated firms. Concavity in enforcement costs generates economies of scale, making it more efficient to target firms with larger tax bases. Concavity in avoidance costs amplifies this mechanism by making larger firms both more responsive to enforcement and less costly to regulate at the margin. These two forces jointly strengthen the case for higher enforcement intensity in high–market power industries, provided market power increases the per-firm tax base.

The design of the tax system further shapes these results. As the system approaches a pure profit tax, the interaction between market power and effective taxation becomes less distortionary, reducing the role of compounded excess burden effects. In this environment, avoidance and enforcement play a relatively more central role in shaping the optimal policy mix.

Finally, the nature of market power itself matters. When market power arises from concentration rather than differences in conduct, market power, firm size, and tax base are more tightly linked. When instead market power reflects softer conduct, these objects can move in different directions, weakening the connection between size-based and market power-based policy rules.

There are several important extensions to potentially explore. For example, this paper assumes fixed levels of competition in each market. However, it is likely that altering tax policy affects market structure through firm entry or exit, mergers and acquisitions, or collusion. If firms merge in response to stricter enforcement in an attempt to exploit opportunities only available to larger sized firms, this may significantly increase the excess burden of enforcement. Endogenizing market structure responses is important to providing a comprehensive illustration of the role of tax policy in addressing market power.

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Appendix

A Firm and Market Responses

In this section, we derive firm and market response margins to each of the tax tools. A given firm's profit maximization problem is

$$\max_{y_k,\gamma_k} y_k q_k(x_k) - C_k(y_k, b_k, \gamma_k, \beta_k) - \tau_k(1-\gamma_k)(y_k q_k - \mu\phi_k(y_k)) \tag{A.1}$$

where costs C are composed of production costs $\phi(y_k)$ and avoidance costs $H(b_k, \gamma_k, \beta_k)$. The first order conditions for production and for avoidance are

$$y_k : (1-(1-\gamma_k)\tau_k - H_b)\left[q_k + \theta_k \frac{\partial q_k}{\partial x_k} y_k\right] = (1-\mu[(1-\gamma_k)\tau_k + H_b])\phi_y \tag{A.2}$$

$$\gamma_k : \tau_k(y_k q_k - \mu\phi_k) = H_\gamma \tag{A.3}$$

where $\theta_k \in [0,1]$ is a conduct parameter that bounds firm responses between perfect competition ($\theta = 0$) and Cournot-Nash ($\theta = 1$). Rearranging the production condition gives the elasticity-adjusted Lerner index

$$\frac{q_k - \frac{1-\mu\omega_k}{1-\omega_k}\phi_y}{q_k}\varepsilon_{x_k} = s_k^i \theta_k \tag{A.4}$$

where $\omega_k = (1-\gamma_k)\tau_k + H_b$ is the operative tax rate faced by the firm, i.e., the tax rate that would lead to the same output decision in a world without tax avoidance, and $s_k^i = \frac{y_k^i}{x_k}$ is the production share of firm i. In the symmetric solution, $s_k^i = \frac{1}{N_k}$. Let $\theta_k^c \equiv s_k \theta_k$ define an industry composite measure of competition equivalent to the Lerner index. Moving forward, q_k' and q_k'' will represent the first and second derivatives of price with respect to quantity.

Lastly, we have that

$$\begin{aligned} q_k + \theta_k^c q_k' x_k - \mu\phi_y &= q_k + \theta_k^c q_k' x_k - \mu\frac{1-\omega_k}{1-\mu\omega_k}\left(q_k + \theta_k^c q_k' x_k\right) \\ &= q_k\left(1 - \frac{\theta_k^c}{\varepsilon_{x_k}}\right)\left[1 - \mu\frac{1-\omega_k}{1-\mu\omega_k}\right] \\ &= q_k\left(1 - \frac{\theta_k^c}{\varepsilon_{x_k}}\right)\left[\frac{1-\mu}{1-\mu\omega_k}\right] \end{aligned} \tag{A.5}$$

Since it must be true that $\varepsilon_{x_k} > \theta_k^c$, this expression must be weakly positive. This is the firm's perceived impact on their size due to a change in the equilibrium quantity. The actual impact on equilibrium firm size is given by

$$\begin{aligned}\frac{db_k^*}{dy_k^*} &= q_k + q_k' N_k y_k - \mu\phi_y \\ &= q_k + \theta_k^c q_k' x_k - \mu\phi_y + q_k' x_k - \theta_k^c q_k' x_k \\ &= q_k\left(\left(1-\frac{\theta_k^c}{\varepsilon_{x_k}}\right)\left[\frac{1-\mu}{1-\mu\omega_k}\right] - \frac{(1-\theta_k^c)}{\varepsilon_{x_k}}\right)\end{aligned} \tag{A.6}$$

Note that this is equal to the firm's internalization of their FOC wedge (previous expression) and the price externality on its rivals, which equals zero for a monopoly. Unlike the previous expression, this is not always weakly positive. For a profit tax ($\mu = 1$), the first term drops out and the expression is weakly negative. For a revenue tax ($\mu = 0$), the bracketed term becomes $\varepsilon_{x_k} - 1$, and the sign depends on whether demand is elastic or inelastic.

A.1 Output, Price, and Size Responses to Tax Tools

To see how each firm and the market responds to changes in the tax tools, we totally differentiate the symmetric FOCs with respect to each tool in turn. We define the following elasticities (demand, demand curvature, and marginal cost):

$$\varepsilon_{x_k} = -x_k' \frac{q_k}{x_k}, \qquad \xi_{x_k} = -x_k \frac{q_k''}{q_k'}, \qquad \eta_k^S = \frac{\phi_{yy}}{\phi_y} y_k \tag{A.7}$$

From the avoidance condition, we also have that

$$\tau_k b_k = H_\gamma \implies \frac{(\tau_k b_k - H_{\gamma b} b_k)}{H_{\gamma\gamma}} = \frac{(H_\gamma - H_{\gamma b} b_k)}{H_{\gamma\gamma}} = \frac{H_\gamma(1-\varepsilon_{H_\gamma^b})}{H_{\gamma\gamma}} = \frac{\gamma_k(1-\varepsilon_{H_\gamma^b})}{\varepsilon_{H_\gamma}} \tag{A.8}$$

where $\varepsilon_{H_\gamma}^r$ is the elasticity of avoidance costs with respect to firm size.

A.1.1 With Respect to the Tax Rate

The change in the per-firm tax base is

$$\frac{db_k}{d\tau_k} = \frac{dy_k}{d\tau_k} q_k + y_k q_k' N_k \frac{dy_k}{d\tau_k} - \mu\phi_y \frac{dy_k}{d\tau_k} = \frac{dy_k}{d\tau_k}[q_k + x_k q_k' - \mu\phi_y] \tag{A.9}$$

Differentiating the production FOC with respect to the tax rate gives

$$\begin{aligned}(1-\omega_k)\left[q_k' N_k \frac{dy_k}{d\tau_k} + \theta_k\left(q_k'' N_k \frac{dy_k}{d\tau_k} y_k + q_k' \frac{dy_k}{d\tau_k}\right)\right] + \left(\frac{d\gamma_k}{d\tau_k}(\tau_k - H_{b\gamma}) - H_{bb}\frac{db_k}{d\tau_k}\right)[q_k + \theta_k q_k' y_k - \mu\phi_y] \\ -(1-\gamma_k)[q_k + \theta_k q_k' y_k - \mu\phi_y] - (1-\mu\omega_k)\phi_{yy}\frac{dy_k}{d\tau_k} = 0\end{aligned} \tag{A.10}$$

Similarly differentiating the avoidance FOC with respect to the tax rate gives

$$b_k + (\tau_k - H_{b\gamma})\frac{db_k}{d\tau_k} = H_{\gamma\gamma}\frac{d\gamma_k}{d\tau_k} \tag{A.11}$$

Plugging this into our previous condition gives

$$\begin{aligned}(1-\omega_k)\frac{dy_k}{d\tau_k}\left(\left[q_k' N_k + \theta_k\left(q_k'' x_k + q_k'\right)\right] + \left(\frac{(\tau_k - H_{b\gamma})^2}{H_{\gamma\gamma}} - H_{bb}\right)[q_k + x_k q_k' - \mu\phi_y][q_k + \theta_k^c q_k' x_k - \mu\phi_y]\right)\\ + \frac{b_k}{H_{\gamma\gamma}}(\tau_k - H_{b\gamma})[q_k + \theta_k^c x_k q_k' - \mu\phi_y] - (1-\gamma_k)[q_k + \theta_k^c q_k' x_k - \phi_y] - (1-\mu\omega_k)\phi_{yy}\frac{dy_k}{d\tau_k} = 0\end{aligned} \tag{A.12}$$

Using the elasticity definitions and the production condition, this can be rearranged as

$$\frac{dy_k}{d\tau_k} = \frac{\left(1 - \gamma_k - \frac{b_k}{H_{\gamma\gamma}}(\tau_k - H_{b\gamma})\right)[q_k + \theta_k^c x_k q_k' - \mu\phi_y]}{(1-\omega_k)N_k q_k'\left[1 + \theta_k^c(1-\xi_{x_k}) + \eta_k^S(\varepsilon_{x_k} - \theta_k^c) + \frac{[q_k + x_k q_k' - \mu\phi_y][q_k + \theta_k^c q_k' x_k - \mu\phi_y]}{N_k q_k'(1-\omega_k)}\left[\frac{(\tau_k - H_{b\gamma})^2}{H_{\gamma\gamma}} - H_{bb}\right]\right]} \tag{A.13}$$

Defining the entire bracketed term in the denominator as Δ, we relabel this expression as

$$\frac{dy_k}{d\tau_k} = \frac{1 - \gamma_k - \frac{b_k}{H_{\gamma\gamma}}(\tau_k - H_{b\gamma})}{(1-\omega_k)N_k q_k'\Delta} q_k\left(1 - \frac{\theta_k^c}{\varepsilon_{x_k}}\right)\left[\frac{1-\mu}{1-\mu\omega_k}\right] \tag{A.14}$$

The change in total output is then simply

$$\frac{dx_k}{d\tau_k} = \frac{1 - \gamma_k - \frac{b_k}{H_{\gamma\gamma}}(\tau_k - H_{b\gamma})}{(1-\omega_k)q_k'\Delta} q_k\left(1 - \frac{\theta_k^c}{\varepsilon_{x_k}}\right)\left[\frac{1-\mu}{1-\mu\omega_k}\right] \tag{A.15}$$

and the semi-elasticity of price with respect to the tax rate, i.e., the pass-through rate, is

$$\rho_k = \frac{1}{q_k}\frac{dq_k}{d\tau_k} = \frac{1 - \gamma_k - \frac{b_k}{H_{\gamma\gamma}}(\tau_k - H_{b\gamma})}{(1-\omega_k)\Delta}\left(1 - \frac{\theta_k^c}{\varepsilon_{x_k}}\right)\left[\frac{1-\mu}{1-\mu\omega_k}\right] \tag{A.16}$$

Note that this is zero under a pure profit tax ($\mu = 1$). Lastly, the change in firm size is

$$\begin{aligned}\frac{db_k}{d\tau_k} &= \frac{1 - \gamma_k - \frac{b_k}{H_{\gamma\gamma}}(\tau_k - H_{b\gamma})}{(1-\omega_k)N_k q_k'\Delta} q_k^2\left(1 - \frac{\theta_k^c}{\varepsilon_{x_k}}\right)\left[\frac{1-\mu}{1-\mu\omega_k}\right]\left(\left(1 - \frac{\theta_k^c}{\varepsilon_{x_k}}\right)\left[\frac{1-\mu}{1-\mu\omega_k}\right] - \frac{(1-\theta_k^c)}{\varepsilon_{x_k}}\right)\\ &= -\frac{1 - \gamma_k - \frac{b_k}{H_{\gamma\gamma}}(\tau_k - H_{b\gamma})}{(1-\omega_k)N_k\Delta} x_k q_k\left(1 - \frac{\theta_k^c}{\varepsilon_{x_k}}\right)\left[\frac{1-\mu}{1-\mu\omega_k}\right]\left((\varepsilon_{x_k} - \theta_k^c)\left[\frac{1-\mu}{1-\mu\omega_k}\right] - (1-\theta_k^c)\right)\end{aligned} \tag{A.17}$$

Again, this is zero under a profit tax. Under CRS avoidance and a revenue tax, this is

$$\frac{db_k}{d\tau_k} = -\frac{1-\gamma_k}{(1-\omega_k)N_k\Delta} x_k q_k\left(\varepsilon_{x_k} - \theta_k^c\right)\left(1 - \frac{1}{\varepsilon_{x_k}}\right) \tag{A.18}$$

This is positive for inelastic demand, and negative for elastic demand.

A.1.2 With Respect to the Enforcement Rate

We follow a similar process for the enforcement rate. Totally differentiating the production condition with respect to enforcement gives

$$(1-\omega_k)\left[q_k' N_k \frac{dy_k}{d\beta_k} + \theta_k\left(q_k'' N_k \frac{dy_k}{d\tau_k} y_k + q_k' \frac{dy_k}{d\beta_k}\right)\right] + \left(\frac{d\gamma_k}{d\beta_k}(\tau_k - H_{b\gamma}) - H_{bb}\frac{db_k}{d\beta_k}\right)[q_k + \theta_k q_k' y_k - \mu\phi_y]$$
$$-H_{b\beta}[q_k + \theta_k q_k' y_k - \mu\phi_y] - (1-\mu\omega_k)\phi_{yy}\frac{dy_k}{d\beta_k} = 0 \tag{A.19}$$

and totally differentiating the avoidance condition gives

$$(\tau_k - H_{b\gamma})\frac{db_k}{d\beta_k} = H_{\gamma\gamma}\frac{d\gamma_k}{d\beta_k} + H_{\gamma\beta} \tag{A.20}$$

Using similar simplifications as before, this leads to the output response

$$\frac{dy_k}{d\beta_k} = \frac{H_{b\beta} - \frac{H_{\gamma\beta}}{H_{\gamma\gamma}}(\tau_k - H_{b\gamma})}{(1-\omega_k)N_k q_k' \Delta} q_k \left(1 - \frac{\theta_k^c}{\varepsilon_{x_k}}\right)\left[\frac{1-\mu}{1-\mu\omega_k}\right] \tag{A.21}$$

which gives total output response

$$\frac{dx_k}{d\beta_k} = \frac{H_{b\beta} - \frac{H_{\gamma\beta}}{H_{\gamma\gamma}}(\tau_k - H_{b\gamma})}{(1-\omega_k) q_k' \Delta} q_k \left(1 - \frac{\theta_k^c}{\varepsilon_{x_k}}\right)\left[\frac{1-\mu}{1-\mu\omega_k}\right] \tag{A.22}$$

and pass-through rate

$$\rho_k^\beta \equiv \frac{1}{q_k}\frac{dq_k}{d\beta_k} = \frac{H_{b\beta} - \frac{H_{\gamma\beta}}{H_{\gamma\gamma}}(\tau_k - H_{b\gamma})}{(1-\omega_k)\Delta}\left(1 - \frac{\theta_k^c}{\varepsilon_{x_k}}\right)\left[\frac{1-\mu}{1-\mu\omega_k}\right] \tag{A.23}$$

Lastly, the change in firm size in response to this enforcement rate

$$\frac{db_k}{d\beta_k} = -\frac{H_{b\beta} - \frac{H_{\gamma\beta}}{H_{\gamma\gamma}}(\tau_k - H_{b\gamma})}{(1-\omega_k)\Delta} x_k q_k \left(1 - \frac{\theta_k^c}{\varepsilon_{x_k}}\right)\left[\frac{1-\mu}{1-\mu\omega_k}\right]\left((\varepsilon_{x_k} - \theta_k^c)\left[\frac{1-\mu}{1-\mu\omega_k}\right] - (1-\theta_k^c)\right) \tag{A.24}$$

Similar to the tax version, this is zero under a pure profit tax. Under a revenue tax with CRS avoidance, this is

$$\frac{db_k}{d\beta_k} = -\frac{H_{b\beta}}{(1-\omega_k)N_k\Delta} x_k q_k (\varepsilon_{x_k} - \theta_k^c)\left(1 - \frac{1}{\varepsilon_{x_k}}\right) \tag{A.25}$$

which similarly can be positive or negative depending on whether the firm is operating in the inelastic or elastic region of demand.

A.2 Output, Price, and Firm Size Responses to Competition

Firm size is defined as the relevant (potential) tax base: $b_k = y_k q_k - \mu\phi(y_k)$. We examine how this changes in response to changes in market power via number of firms and via conduct.

A.2.1 Changes with Respect to the Number of Firms

The change in equilibrium firm size as the number of firms increases is given by

$$\begin{aligned}\frac{db_k}{dN_k} &= \frac{dy_k}{dN_k}q_k + y_k q_k'\left(N_k\frac{dy_k}{dN_k} + y_k\right) - \mu\phi_y\frac{dy_k}{dN_k} \\ &= \frac{dy_k}{dN_k}\left[q_k + x_k q_k' - \mu\phi_y\right] + y_k^2 q_k' \\ &= \frac{dy_k}{dN_k}x_k q_k'\left[1 - \theta_k^c - \frac{1-\mu}{1-\mu\omega_k}(\varepsilon_{x_k} - \theta_k^c)\right] + y_k^2 q_k'\end{aligned} \tag{A.26}$$

Starting with the production FOC, we totally differentiate with respect to the number of firms:

$$\begin{aligned}(1-\omega_k)\left[q_k' N_k\frac{dy_k}{dN_k} + \theta_k q_k'' N_k y_k\frac{dy_k}{dN_k}\right] + \left(\frac{d\gamma_k}{dN_k}(\tau_k - H_{b\gamma}) - H_{bb}\frac{db_k}{dN_k}\right)\left[q_k + \theta_k^c q_k' x_k - \mu\phi_y\right] \\ +(1-\omega_k)\left[q_k' y_k + \theta_k q_k'' y_k^2\right] - (1-\mu\omega_k)\phi_{yy}\frac{dy_k}{dN_k} = 0\end{aligned} \tag{A.27}$$

Similarly differentiating the avoidance FOC gives $\tau_k\frac{db_k}{dN_k} = H_{bb}\frac{db_k}{dN_k} + H_{\gamma\gamma}\frac{d\gamma_k}{dN_k}$. Plugging this in for the avoidance response gives

$$\begin{aligned}(1-\omega_k)\left[q_k' N_k\frac{dy_k}{dN_k} + \theta_k q_k'' x_k\frac{dy_k}{dN_k}\right] + \left(\frac{db_k}{dN_k}\left[\frac{(\tau_k - H_{b\gamma})^2}{H_{\gamma\gamma}} - H_{bb}\right]\right)\left[q_k + \theta_k^c q_k' x_k - \mu\phi_y\right] \\ +(1-\omega_k)\left[q_k' y_k + \theta_k q_k'' y_k^2\right] - (1-\mu\omega_k)\phi_{yy}\frac{dy_k}{dN_k} = 0\end{aligned} \tag{A.28}$$

Lastly, we plug in the first expression relating firm size and firm output to get

$$\begin{aligned}\frac{dy_k}{dN_k}(1-\omega_k)q_k' N_k\left(1 + \theta_k^c(1-\xi_{x_k}) + \eta_k^S(\varepsilon_{x_k} - \theta_k^c) + \frac{\frac{db_k^*}{dy_k^*}[q_k + \theta_k^c q_k' x_k - \mu\phi_y]}{N_k q_k'(1-\omega_k)}\left[\frac{(\tau_k - H_{b\gamma})^2}{H_{\gamma\gamma}} - H_{bb}\right]\right) \\ = -(1-\omega_k)\left[q_k' y_k + \theta_k q_k'' y_k^2 + \frac{y_k^2 q_k'}{1-\omega_k}[q_k + \theta_k^c q_k' x_k - \mu\phi_y]\left[\frac{(\tau_k - H_{b\gamma})^2}{H_{\gamma\gamma}} - H_{bb}\right]\right]\end{aligned} \tag{A.29}$$

$$\begin{aligned}\frac{dy_k}{dN_k}(1-\omega_k)q_k' N_k\left(1 + \theta_k^c(1-\xi_{x_k}) + \eta_k^S(\varepsilon_{x_k} - \theta_k^c) + \frac{\frac{db_k^*}{dy_k^*}[q_k + \theta_k^c q_k' x_k - \mu\phi_y]}{N_k q_k'(1-\omega_k)}\left[\frac{(\tau_k - H_{b\gamma})^2}{H_{\gamma\gamma}} - H_{bb}\right]\right) \\ = -(1-\omega_k)q_k' y_k\left[1 - \theta_k^c\xi_{x_k} + \frac{y_k}{1-\omega_k}[q_k + \theta_k^c q_k' x_k - \mu\phi_y]\left[\frac{(\tau_k - H_{b\gamma})^2}{H_{\gamma\gamma}} - H_{bb}\right]\right]\end{aligned} \tag{A.30}$$

Thus, the change in per-firm output is given by

$$\frac{dy_k}{dN_k} = -\frac{1 - \theta_k^c\xi_{x_k} + \frac{y_k}{1-\omega_k}[q_k + \theta_k^c q_k' x_k - \mu\phi_y]\left[\frac{(\tau_k - H_{b\gamma})^2}{H_{\gamma\gamma}} - H_{bb}\right]}{\Delta}\frac{y_k}{N_k} < 0 \tag{A.31}$$

Since $\frac{dx_k}{dN_k} = y_k + N_k \frac{dy_k}{dN_k}$, this gives the total change in output as

$$\frac{dx_k}{dN_k} = y_k \left[1 - \frac{1 - \theta_k^c \xi_{x_k} + \frac{y_k}{1-\omega_k}[q_k + \theta_k^c q_k' x_k - \mu\phi_y]\left[\frac{(\tau_k - H_{b\gamma})^2}{H_{\gamma\gamma}} - H_{bb}\right]}{\Delta}\right]$$
$$= y_k \frac{\theta_k^c + \eta_k^S(\varepsilon_{x_k} - \theta_k^c) + \left(\frac{y_k}{1-\omega_k} - \frac{\frac{db_k^*}{dy_k^*}}{N_k q_k'(1-\omega_k)}\right)[q_k + \theta_k^c q_k' x_k - \mu\phi_y]\left[\frac{(\tau_k - H_{b\gamma})^2}{H_{\gamma\gamma}} - H_{bb}\right]}{\Delta} > 0 \tag{A.32}$$

Finally, the change in the real tax base per firm can be written

$$\frac{db_k}{dN_k} = \frac{dy_k}{dN_k}(q_k - \mu\phi_y) + y_k q_k' \frac{dx_k}{dN_k} < 0 \tag{A.33}$$

where the inequality holds since we showed $\frac{dy_k}{dN_k} < 0$ and $\frac{dx_k}{dN_k} > 0$. Price must be above marginal cost, and and $q_k' < 0$ by the law of demand. If we want to express this in terms of a change in the index θ_k^c, this simply follows as

$$\left.\frac{db_k}{d\theta_k^c}\right|_{dN_k} = \frac{db_k}{dN_k}\left[\frac{1}{d\theta_k^c/dN_k}\right] = \frac{db_k}{dN_k}\left[\frac{-N_k}{\theta_k^c}\right] > 0 \tag{A.34}$$

Under CRS avoidance, the output response simplifies to

$$\frac{dx_k}{dN_k} = \frac{\theta_k^c + \eta_k(\varepsilon_x - \theta_k^c)}{\Delta} y_k \tag{A.35}$$

The price response is

$$\frac{dq_k}{dN_k} = \frac{\theta_k^c + \eta_k(\varepsilon_x - \theta_k^c)}{\Delta} q_k' y_k \tag{A.36}$$

And in terms of the composite index, we have that

$$\left.\frac{dy_k}{dN_k}\right|_{dn} = \frac{dy_k}{dN_k}\left[\frac{1}{d\theta_k^c/dN_k}\right] = \frac{1 - \theta_k^c \xi_k}{\theta_k^c} \tag{A.37}$$

A.2.2 Changes with Respect to Conduct

The change in the real tax base per-firm in response to softer conduct is:

$$\begin{aligned}\frac{db_k}{d\theta_k} &= \frac{dy_k}{d\theta_k} q_k + y_k q_k' \left(N_k \frac{dy_k}{d\theta_k}\right) - \mu\phi_y \frac{dy_k}{d\theta_k} \\ &= \frac{dy_k}{d\theta_k} x_k q_k' \left[1 - \theta_k^c - \frac{1-\mu}{1-\mu\omega_k}(\varepsilon_{x_k} - \theta_k^c)\right]\end{aligned} \tag{A.38}$$

Differentiating the symmetric production FOC with respect to θ gives (after plugging in the avoidance side: $\tau_k \frac{db_k}{d\theta_k} = H_{\gamma\gamma} \frac{d\gamma_k}{d\theta_k} + H_{b\gamma} \frac{db_k}{d\theta_k}$)

$$(1-\omega_k)\left[q_k' N_k \frac{dy_k}{d\theta_k} + \theta_k q_k'' N_k y_k \frac{dy_k}{d\theta_k}\right] + \left(\frac{db_k}{d\theta_k}\left[\frac{(\tau_k - H_{b\gamma})^2}{H_{\gamma\gamma}} - H_{bb}\right]\right)\left[q_k + \theta_k^c q_k' x_k - \mu\phi_y\right] \\ +(1-\omega_k) q_k' y_k - (1-\mu\omega_k)\phi_{yy} \frac{dy_k}{d\theta_k} = 0 \tag{A.39}$$

Plugging in our first expression and then simplifying as before gives the change in per-firm output as

$$\frac{dy_k}{d\theta_k} = -\frac{1}{\Delta}\frac{y_k}{N_k} < 0 \tag{A.40}$$

which implies a total output response of

$$\frac{dx_k}{d\theta_k} = -\frac{1}{\Delta} y_k < 0 \tag{A.41}$$

and a real per-firm tax base response

$$\frac{db_k}{d\theta_k} = \underbrace{-\frac{1}{\Delta} y_k^2 q_k'}_{>0} \left[1 - \theta_k^c - \frac{1-\mu}{1-\mu\omega_k}(\varepsilon_{x_k} - \theta_k^c)\right] \tag{A.42}$$

This expression does *not* have an unambiguous sign. Depending on the demand elasticity of the industry, it may be the case that softer conduct increases or decreases the per-firm tax base. For example, under a pure revenue tax ($\mu = 0$), this expression becomes

$$\frac{db_k}{d\theta_k} = \underbrace{-\frac{1}{\Delta} y_k^2 q_k'}_{>0} \left[1 - \varepsilon_{x_k}\right] \tag{A.43}$$

which is positive if $\varepsilon_{x_k} < 1$ and negative if $\varepsilon_{x_k} > 1$. Meanwhile, under a pure profit tax, the expression simplifies to

$$\frac{db_k}{d\theta_k} = \underbrace{-\frac{1}{\Delta} y_k^2 q_k'}_{>0} \left[1 - \theta_k^c\right] \tag{A.44}$$

which is positive so long as $\theta_k^c < 1$ (i.e., there is room for market power to increase). Thus, as expected, firm size (profits) unambiguously increases with softer conduct.

A.3 Profit Responses

Differentiating the firm's after tax profits $\pi_k = y_k q_k - C(\cdot) - \tau_k(1-\gamma_k)(y_k q_k - \mu\phi_k)$ with respect to the tax rate gives

$$\frac{\partial \pi_k}{\partial \tau_k} = (1-H_b)\left[\frac{\partial y_k}{\partial \tau_k} q_k + y_k \frac{\partial q_k}{\partial \tau_k}\right] - \phi_y \frac{\partial y_k}{\partial \tau_k} + H_b \mu \phi_y \frac{\partial y_k}{\partial \tau_k} - H_\gamma \frac{\partial \gamma_k}{\partial \tau_k} - z_k$$
$$- \tau_k \left[-\frac{\partial \gamma_k}{\partial \tau_k}(y_k q_k - \mu\phi_k) + (1-\gamma_k)\left[\frac{\partial y_k}{\partial \tau_k} q_k + y_k \frac{\partial q_k}{\partial \tau_k} - \mu\phi_y \frac{\partial y_k}{\partial \tau_k}\right]\right]$$

where z_k is the firm's reported taxable income. This can be rearranged as

$$\frac{\partial \pi_k}{\partial \tau_k} = -z_k + (1-\omega_k)\left[\frac{\partial y_k}{\partial \tau_k} q_k + y_k \frac{\partial q_k}{\partial \tau_k}\right] - (1-\mu\omega_k)\phi_y \frac{\partial y_k}{\partial \tau_k} + (\tau_k b_k - H_\gamma)\frac{\partial \gamma_k}{\partial \tau_k}$$

Plugging in the first order for production and avoidance gives

$$\frac{\partial \pi_k}{\partial \tau_k} = -z_k + (1-\omega_k) y_k q_k' \left[\frac{\partial x_k}{\partial \tau_k} - \theta \frac{\partial y_k}{\partial \tau_k}\right]$$

Using the market clearing condition $x_k = N_k y_k$ then gives the final expression

$$\frac{\partial \pi_k}{\partial \tau_k} = -z_k + (1-\omega_k)\frac{N_k - \theta}{N_k}\frac{\partial q_k}{\partial \tau_k} y_k$$

or

$$\frac{\partial \pi_k}{\partial \tau_k} = -z_k + (1-\omega_k)(1-\theta_k^c)\frac{\partial q_k}{\partial \tau_k} y_k$$

which is the direct effect of the tax plus the competitive response. Similarly, a firm's profit change due to an increase in enforcement is given by

$$\frac{\partial \pi_k}{\partial \beta_k} = -H_\beta + (1-\omega_k)(1-\theta_k^c)\frac{\partial q_k}{\partial \beta_k} y_k$$

The aggregate industry profit (Π_k) responses are then given by

$$\frac{\partial \Pi_k}{\partial \tau_k} = -Z_k + (1-\omega_k)(1-\theta_k^c)\frac{\partial q_k}{\partial \tau_k} x_k$$

$$\frac{\partial \Pi_k}{\partial \beta_k} = -N_k H_\beta + (1-\omega_k)(1-\theta_k^c)\frac{\partial q_k}{\partial \beta_k} x_k$$

Under a profit tax, only the direct effect on profits remains. Under a revenue tax with CRS avoidance, these responses become

$$\begin{aligned}\frac{\partial \Pi_k}{\partial \tau_k} &= -Z_k + (1-\theta_k^c)\frac{(1-\gamma_k)x_k q_k}{\Delta}\left(1-\frac{\theta_k^c}{\varepsilon_{x_k}}\right) \\ &= -Z_k\left[1-\frac{(1-\theta_k^c)\left(1-\frac{\theta_k^c}{\varepsilon_{x_k}}\right)}{\Delta}\right]\end{aligned} \tag{A.45}$$

$$\frac{\partial \Pi_k}{\partial \beta_k} = -H_\beta\left[N_k - \frac{(1-\theta_k^c)\left(1-\frac{\theta_k^c}{\varepsilon_{x_k}}\right)}{\Delta}x_k q_k\right] \tag{A.46}$$

A.4 Welfare Effect

The effect of a change in one of the tax tools on welfare is the effect on the representative consumer, i.e.,

$$\frac{dW}{dT_k} = \alpha\left[-\frac{\partial q_k}{\partial T_k}x_k + \frac{\partial \Pi_k}{\partial T_k}\right]$$

where $T_k \in \{\tau_k, \beta_k\}$. Starting with the tax rate, plugging in the value for the profit change calculated previously gives

$$\frac{1}{\alpha Z_k}\frac{dW}{d\tau_k} = -\left[1+\frac{\partial q_k}{\partial \tau_k}\frac{x_k}{Z_k}\left[1-(1-\omega_k)(1-\theta_k^c)\right]\right] = -\left[1+\frac{\partial q_k}{\partial \tau_k}\frac{x_k}{Z_k}\left[\omega_k+\theta_k^c(1-\omega_k)\right]\right]$$

Define the "1" as the mechanical dollar raised, while the rest of the expression is the excess burden. Thus, this expression simplifies to

$$\frac{1}{\alpha Z_k}\frac{dW}{d\tau_k} = -\left[1+EB_k\right]$$

Similarly, for the enforcement, the welfare effect is

$$\frac{1}{\alpha Z_k}\frac{dW}{d\beta_k} = -\left[\frac{N_k H_\beta}{Z_k}+\frac{\partial q_k}{\partial \beta_k}\frac{x_k}{Z_k}\left[1-(1-\omega_k)(1-\theta_k^c)\right]\right] = -\left[\frac{N_k H_\beta}{Z_k}+\frac{\partial q_k}{\partial \beta_k}\frac{x_k}{Z_k}\left[\omega_k+\theta_k^c(1-\omega_k)\right]\right]$$

Note that for enforcement, this entire expression is excess burden as there is no mechanical transfer. The government only raises revenue if the firm(s) behaviorally respond to the change in the enforcement rate.

$$\frac{1}{\alpha Z_k}\frac{dW}{d\beta_k} = -EB_k^\beta$$

Under a pure profit tax where $\mu = 1$, the price pass-through is zero for both tools. In this case, there is no excess burden of taxation ($EB_k = 0$), but there is still burden for the enforcement

$\left(EB_k^{\beta} = \frac{N_k H_\beta}{Z_k}\right)$ due to increased compliance cost. Thus, the per taxable income welfare effect of taxation is independent of competition (and firm size). Meanwhile, the welfare effect of enforcement may still have a dependency on market power as $\frac{H_\beta}{z_k}$ is dependent on the avoidance rate. This relationship is discussed more in Lemma 2.1.

A.5 Taxable Income Effect

A.5.1 With Respect to the Tax Rate

The relevant elasticity is the elasticity of taxable income, which is reported revenue for an output tax ($\mu = 0$) case and reported production profits for a profit tax case ($\mu = 1$). Then the elasticity is

$$\frac{\varepsilon_{Z_k}}{1-\tau_k} \equiv \frac{1}{Z_k}\frac{\partial Z_k}{\partial(1-\tau_k)} = \frac{\partial(1-\gamma_k)}{\partial(1-\tau_k)}\frac{B_k}{Z_k} - \frac{(1-\gamma_k)}{Z_k}\frac{\partial B_k}{\partial \tau_k}$$

Since, under CRS avoidance, we have that

$$\frac{\partial R}{\partial \tau_k} = -\left((\varepsilon_{x_k} - \theta_k^c)\left[\frac{1-\mu}{1-\mu\omega_k}\right] - (1-\theta_k^c)\right)\frac{(1-\gamma_k)x_k q_k}{(1-\omega_k)\Delta}\left(1-\frac{\theta_k^c}{\varepsilon_{x_k}}\right)\left[\frac{1-\mu}{1-\mu\omega_k}\right] \tag{A.47}$$

Under a profit tax, this entire expression is zero. This implies that the elasticity of reported taxable income is exactly equal to the tax elasticity of the avoidance rate. Under a revenue tax with CRS avoidance, this becomes

$$\frac{\partial B_k}{\partial \tau_k} = (1-\varepsilon_{x_k})\frac{(1-\gamma_k)x_k q_k}{(1-\omega_k)\Delta}\left(1-\frac{\theta_k^c}{\varepsilon_{x_k}}\right) \tag{A.48}$$

and the elasticity of taxable income is

$$\varepsilon_{Z_k} = \varepsilon_{1-\gamma_k} - \frac{(1-\tau_k)(1-\gamma_k)}{1-\omega_k}\frac{(1-\varepsilon_{x_k})}{\Delta}\left(1-\frac{\theta_k^c}{\varepsilon_{x_k}}\right) \tag{A.49}$$

A.5.2 With Respect to the Enforcement Intensity

Smilarly, the elasticity of enforcement is defined by

$$\frac{\varepsilon_{Z_k}^{\beta_k}}{\beta_k} \equiv \frac{1}{Z_k}\frac{\partial Z_k}{\partial \beta_k} = \frac{\partial(1-\gamma_k)}{\partial \beta_k}\frac{B_k}{Z_k} + \frac{(1-\gamma_k)}{Z_k}\frac{\partial B_k}{\partial \beta_k}$$

where

$$\frac{\partial R}{\partial \beta_k} = -\left((\varepsilon_{x_k} - \theta_k^c)\left[\frac{1-\mu}{1-\mu\omega_k}\right] - (1-\theta_k^c)\right)\frac{H_{b\beta} x_k q_k}{(1-\omega_k)\Delta}\left(1-\frac{\theta_k^c}{\varepsilon_{x_k}}\right)\left[\frac{1-\mu}{1-\mu\omega_k}\right] \tag{A.50}$$

As before this is zero under a profit tax and under an output tax becomes

$$\frac{\partial B_k}{\partial \beta_k} = (1 - \varepsilon_{x_k}) \frac{H_{b\beta} x_k q_k}{(1-\omega_k)\Delta} \left(1 - \frac{\theta_k^c}{\varepsilon_{x_k}}\right) \tag{A.51}$$

and the elasticity of taxable income is

$$\varepsilon^{\beta}_{Z_k} = \varepsilon^{\beta}_{1-\gamma_k} + \frac{\beta_k H_{b\beta}}{1-\omega_k} \frac{(1-\varepsilon_{x_k})}{\Delta} \left(1 - \frac{\theta_k^c}{\varepsilon_{x_k}}\right) \tag{A.52}$$

B Proofs

B.1 Lemma 1

By assumption, the avoidance cost function is multiplicatively separable in its three arguments:

$$H(b_k, \gamma_k, \beta_k) = G(b_k)\, I(\gamma_k)\, J(\beta_k) \tag{A.53}$$

G, I, and J are positive, continuously differentiable, and satisfy the following assumptions: $I_\gamma > 0$, $I_{\gamma\gamma} > 0$, $G(r) > 0$ for $r > 0$, and $G(0) = 0$. Using the avoidance FOC, he optimal avoidance rate satisfies

$$I_\gamma(\gamma_k) = \frac{b_k \tau_k}{G(b_k) J(\beta)}$$

Since $I_{\gamma\gamma} > 0$, I_γ is strictly increasing and therefore has a strictly increasing inverse I_γ^{-1}. This implies that

$$\gamma_k^* = I_\gamma^{-1}\left(\frac{b_k \tau_k}{G(b_k) J(\beta)}\right)$$

Firm Size and Avoidance Since I_γ^{-1} is strictly increasing, the sign of $\frac{\partial \gamma_k^*}{\partial b_k}$ is determined by the sign of

$$\frac{d}{db_k}\left[\frac{b_k}{G(b_k)}\right] = \frac{G(b_k) - b_k\, G'(b_k)}{G(b_k)^2} = \frac{1}{G(b_k)}\left(1 - \frac{b_k\, G'(b_k)}{G(b_k)}\right) = \frac{1}{G(b_k)}\Big(1 - \varepsilon_H^b(b_k)\Big) \tag{A.54}$$

where $\varepsilon_H^b \equiv \frac{b_k G'(b_k)}{G(b_k)} = \frac{b_k H_b}{H}$ is the elasticity of H with respect to b. Since $G(b_k) > 0$, we have

$$\operatorname{sign}\left(\frac{\partial \gamma_k^*}{\partial b_k}\right) = \operatorname{sign}\Big(1 - \varepsilon_H^b\Big) \tag{A.55}$$

Therefore, for a given tax rate τ and enforcement intensity β,

- If $\varepsilon_H^b < 1$ (H is strictly concave in b), larger firms choose *higher* avoidance rates: $\frac{\partial \gamma^*}{\partial r} > 0$.
- If $\varepsilon_H^b > 1$ (H is strictly convex in b), larger firms choose *lower* avoidance rates: $\frac{\partial \gamma^*}{\partial r} < 0$.

- If $\varepsilon_H^b = 1$ (H is linear in b), the avoidance rate is independent of firm size.

Market Power and Avoidance. For a change in the number of firms, we have

$$\frac{d\gamma_k^*}{dN_k} = \underbrace{\frac{d\gamma_k}{db_k}}_{>0} \cdot \underbrace{\frac{db_k}{dN_k}}_{<0} \tag{A.56}$$

which implies that

$$\text{sign}\left(\frac{d\gamma_k^*}{dN_k}\right) = -\,\text{sign}\left(1 - \varepsilon_H^b\right) \tag{A.57}$$

For softer conduct (θ), we have

$$\frac{d\gamma_k^*}{d\theta_k} = \underbrace{\frac{d\gamma_k}{db_k}}_{>0} \cdot \frac{db_k}{d\theta_k} \tag{A.58}$$

This sign of $\frac{db_k}{d\theta_k}$ can be ambiguous and depends on the cost deductibility μ. For $\mu = 1$, and the real tax base is equal to production profits, then softer conduct increases firm size, $\frac{db_k}{d\theta_k} > 0$. The change in optimal avoidance follows the same sign as decreasing N_k. For $\mu = 0$, the real tax base is equal to firm revenue. If conduct softens under inelastic demand, revenue per firm increases and avoidance follows the same direction as above. Under elastic demand, revenue per firm decreases, and avoidance follows the opposite direction as above.

B.2 Lemma 2

Under the CRS avoidance, the pass-through rate is given by

$$\rho_k \equiv \frac{1}{q_k}\frac{dq_k}{d\tau_k} = \left[\frac{1-\mu}{1-\mu\omega_k}\right]\left(1 - \frac{\theta_k^c}{\varepsilon_{x_k}}\right)\frac{1-\gamma_k}{(1-\omega_k)\Delta_k} \tag{A.59}$$

where

$$\Delta_k = 1 + \theta_k^c(1-\xi_{x_k}) + \eta_{s_k}(\varepsilon_k - \theta_k^c) \tag{A.60}$$

Suppose $(\varepsilon_x, \eta_s, \xi_x)$ is held fixed across two markets, where the second market is marginally less competitive than the other. Define $f(\theta) = \left(1 - \frac{\theta_k^c}{\varepsilon_{x_k}}\right)\frac{1}{\Delta_k}$. Holding the previous elasticities fixed, then

$$\frac{df(\theta)}{d\theta} = \frac{-\frac{1}{\varepsilon_x}\Delta - \left(1 - \frac{\theta_k^c}{\varepsilon_{x_k}}\right)(1 - \xi_x - \eta_s)}{\Delta^2} \tag{A.61}$$

This is positive if and only if

$$-\frac{1}{\varepsilon_{x_k}} - \frac{\theta_k^c}{\varepsilon_x}(1-\xi_x) - \eta_s\left(1 - \frac{\theta_k^c}{\varepsilon_x}\right) - \left(1 - \frac{\theta_k^c}{\varepsilon_{x_k}}\right)(1 - \xi_x - \eta_s) > 0 \tag{A.62}$$

or, simplified down,

$$\xi_x - \frac{1}{\varepsilon_x} > 1 \tag{A.63}$$

Thus, we've shown how the Ritz (2024) result for specific taxation does not translate directly to revenue taxation (ad valorem). The difference is in the markup adjustment factor $\left(1-\frac{\theta_k^c}{\varepsilon_{x_k}}\right)$. Otherwise, the condition that remains would be the condition for specific taxation which is $\xi_x+\eta_s>1$, which includes the firm supply term. Thus, it is only possible for revenue tax pass-through to be increasing with market power if demand is log-convex. In most common cases (log-concave), pass-through decreases with market power (controlling for demand, curvature, and supply elasticities).

B.2.1 Excess Burden of Taxation with Competition

Now we examine how the excess burden term changes as the degree of competition changes. Under a profit tax excess burden is equal to 0 (since price is unresponsive) regardless of the competition. Remember that, under constant scale avoidance technology, the pass-through is

$$\rho_k=\left(1-\frac{\theta_k^c}{\varepsilon_{x_k}}\right)\frac{1-\gamma_k}{(1-\omega_k)\Delta}\left[\frac{1-\mu}{1-\mu\omega_k}\right] \tag{A.64}$$

Since

$$EB_k=\rho_k\frac{x_kq_k}{Z_k}\left[\omega_k+(1-\omega_k)\,\theta_k^c\right]$$

Under a revenue tax, this becomes

$$\begin{aligned}EB_k&=\left(1-\frac{\theta_k^c}{\varepsilon_{x_k}}\right)\frac{(1-\gamma_k)x_kq_k}{Z_k}\frac{\left[1-(1-\omega_k)\,(1-\theta_k^c)\right]}{1-\omega_k}\frac{1}{\Delta}\\&=\left(1-\frac{\theta_k^c}{\varepsilon_{x_k}}\right)\frac{\left[1-(1-\omega_k)\,(1-\theta_k^c)\right]}{1-\omega_k}\frac{1}{1+\theta_k^c(1-\xi_k)+\eta_k^S(\varepsilon_{x_k}-\theta_k^c)}\end{aligned} \tag{A.65}$$

Consider a between markets comparison, where $(\varepsilon_x,\xi_x,\eta_s)$ are equal across the markets being compared. Under our previous notation, we have that

$$EB_k=\frac{\omega_k+(1-\omega_k)\theta_k^c}{1-\omega_k}f(\theta_k^c) \tag{A.66}$$

Then

$$\begin{aligned}\frac{dEB_k}{d\theta_k^c}&=f(\theta_k^c)+\frac{\omega_k+(1-\omega_k)\theta_k^c}{1-\omega_k}f'(\theta_k^c)\\&=\frac{1}{\Delta^2}\left[\left(1-\frac{\theta_k^c}{\varepsilon_{x_k}}\right)\Delta-\frac{\omega_k+(1-\omega_k)\theta_k^c}{1-\omega_k}\left(1-\xi_{x_k}+\frac{1}{\varepsilon_{x_k}}\right)\right]\end{aligned} \tag{A.67}$$

This is positive if and only if

$$\left(1-\frac{\theta_k^c}{\varepsilon_{x_k}}\right)\Delta>\frac{\omega_k+(1-\omega_k)\theta_k^c}{1-\omega_k}\left(1-\xi_{x_k}+\frac{1}{\varepsilon_{x_k}}\right) \tag{A.68}$$

Note that $1-\xi_{x_k}+\frac{1}{\varepsilon_{x_k}}=0$ under isoelastic demand. Thus, for demand at least as convex as isoelastic demand, the marginal excess burden of taxation is always increasing with market power.

We can see that as $\theta \to 0$ and $\omega \to 0$, the RHS is also close to 0. Thus, while excess burden is not necessarily monotonic with market power, we can state that starting near 0, there is positive excess burden in imperfect markets compared to perfect competition.

For a within markets approach, we perform the same exercise but do not hold the elasticities constant. As this has significantly more moving parts, we consider only a simplified version where marginal costs are constant ($\eta^S = 0$), and demand is linear ($\xi_x = 0$). The excess burden simplifies to

$$\left(1 - \frac{\theta_k^c}{\varepsilon_{x_k}}\right) \frac{\omega_k + (1-\omega_k)\theta_k^c}{1-\omega_k} \frac{1}{1+\theta_k^c} \tag{A.69}$$

Under perfect competition ($\theta_k^c = 0$), the excess burden becomes

$$EB_{PC} = \frac{\omega_k}{1-\omega_k} \frac{1}{1+\eta_k^S \varepsilon_{x_k}} \tag{A.70}$$

while under a monopoly, this becomes

$$EB_M = \frac{1 - \frac{1}{\varepsilon_{x_k}}}{1-\omega_k} \frac{1}{1 + (1-\xi_k) + \eta_k^S(\varepsilon_{x_k} - 1)} \tag{A.71}$$

Note that, under CRS avoidance, $\omega_k = (1-\gamma_k)\tau_k + H(1, \gamma, \beta)$, which is independent of firm size. Note that, starting at zero taxation and enforcement, the monopoly unambiguously has a higher excess burden from introducing effective taxation. For a nonzero tax, the monopoly excess burden is larger if the (operative) tax rate is not too high. For example, consider a linear demand and constant marginal cost example (the two industries need different marginal costs in order to match on observed elasticities). Then $EB_{PC} = \frac{\omega_k}{1-\omega_k}$ and $EB_M = \frac{1-\frac{1}{\varepsilon_{x_k}}}{1-\omega_k}\frac{1}{2}$. The the excess burden is higher under the monopoly if and only if $\omega < \frac{1}{2}\left(1 - \frac{1}{\varepsilon_x}\right)$.

B.2.2 Compliance and Excess Burden of Enforcement with Competition

Under a profit tax, the real compliance costs are

$$\frac{N_k H_\beta}{Z_k} = \frac{1}{1-\gamma_k} \frac{H_\beta}{b_k} \tag{A.72}$$

Using multiplicative separability,

$$\frac{H_\beta}{b_k} \propto \frac{G(b_k)}{b_k} I\left(I_\gamma^{-1}\left(\frac{b_k}{G(b_k)}\right)\right) \tag{A.73}$$

Using the change of variable $u = \frac{b_k}{G(b_k)}$, we have that

$$\begin{aligned}\frac{d\ln\left(\frac{H_\beta}{b_k}\right)}{d\ln(u)} &\propto \frac{d\ln(1/u)}{d\ln(u)} + \frac{d\ln\left(I\left(I_\gamma^{-1}(u)\right)\right)}{d\ln(u)} \\ &= -1 + \frac{I_\gamma^2(\gamma_k)}{I(\gamma_k)I_{\gamma\gamma}(\gamma_k)}\end{aligned} \tag{A.74}$$

This is negative as long as I is log-concave in γ, which is the already assumed regularity condition. Thus, $\frac{H_\beta}{b_k}$ is decreasing in u, which means that

$$\frac{N_k H_\beta}{Z_k} = \frac{1}{1-\gamma_k}\frac{H_\beta}{b_k} \propto \frac{G(b_k)}{b_k} \tag{A.75}$$

Thus compliance costs are increasing (decreasing) in b_k if G is convex (concave). The relationships with market power follow from Lemma 3.

Under CRS avoidance, this is

$$EB_k^\beta = \frac{H_\beta(1,\gamma,\beta)}{1-\gamma_k} \tag{A.76}$$

which is independent of firm level differences and competition since γ will be a function of only the tax and enforcement rate. Under IRS avoidance, $\frac{d\gamma_k}{d\theta_k} > 0$, $\frac{db_k}{d\theta_k} > 0$, and $H_\beta - b_k H_{\beta b}$. Under DRS avoidance, both effects go in the same direction, a Under a revenue tax, this expands to

$$\begin{aligned}EB_k^\beta &= \frac{H_\beta(1,\gamma,\beta)}{1-\gamma_k} + \frac{H_{r\beta}}{1-\gamma_k}\left(1 - \frac{\theta_k^c}{\varepsilon_{x_k}}\right)\frac{\left[1-(1-\omega_k)\left(1-\theta_k^c\right)\right]}{1-\omega_k}\frac{1}{\Delta} \\ &= \frac{H_\beta(1,\gamma,\beta)}{1-\gamma_k}\left[1 + \frac{\omega_k + (1-\omega_k)\theta_k^c}{1-\omega_k} f(\theta_k^c)\right]\end{aligned} \tag{A.77}$$

The new terms here only shift the levels of the excess burden in comparison to the tax expression. Thus, the relationship between excess burden of enforcement and the excess burden of taxation versus market power follow the same condition as the tax expression.

B.3 Lemma 3 and Proposition 1

Consider the reported tax base of the industry $Z_k = (1-\gamma_k)(x_k q_k - \mu N_k \phi(y_k))$. The change in this with respect to a tax tool T_k is $\frac{\partial Z_k}{\partial T_k} = \frac{\partial(1-\gamma_k)}{\partial T_k} B_k + (1-\gamma_k)\frac{\partial B_k}{\partial T_k}$.

Tax: For the tax rate, the reported taxable income response is

$$\begin{aligned}\frac{\partial Z_k}{\partial(1-\tau_k)} &= \frac{1}{H_{\gamma\gamma}}\left[b_k - (\tau_k - H_{b\gamma})\frac{\partial b_k}{\partial \tau_k}\right] B_k - (1-\gamma_k)\frac{\partial B_k}{\partial \tau_k} \\ &= \frac{b_k}{H_{\gamma\gamma}} B_k - \left(1 - \gamma_k + \frac{(\tau_k - H_{b\gamma})b_k}{H_{\gamma\gamma}}\right)\frac{\partial B_k}{\partial \tau_k}\end{aligned} \tag{A.78}$$

Under a revenue tax and CRS avoidance technology,

$$\varepsilon_{Z_k} = \frac{b_k}{H_{\gamma\gamma}}\frac{1-\tau_k}{1-\gamma_k} - \frac{(1-\tau_k)(1-\gamma_k)}{1-\omega_k}\frac{(1-\varepsilon_{x_k})}{\Delta}\left(1-\frac{\theta^c_k}{\varepsilon_{x_k}}\right) \tag{A.79}$$

The avoidance rate is independent of the competition index (through either firm number or conduct). Thus,

$$\frac{d\varepsilon_{Z_k}}{d\theta^c_k} \propto -(1-\varepsilon_{x_k})\frac{df(\theta^c_k)}{d\theta^c_k} + \frac{d\varepsilon_{x_k}}{d\theta^c_k}f(\theta^c_k) \tag{A.80}$$

where $f(\theta^c_k)$ is defined similarly to Lemma 2. In a cross-market comparison where ε_x is fixed, then the direction of $\frac{d\varepsilon_{Z_k}}{d\theta^c_k}$ depends on whether demand is elastic or not.

Under a profit tax,

$$\varepsilon_{Z_k} = \frac{b_k}{H_{\gamma\gamma}}\frac{1-\tau_k}{1-\gamma_k} = \frac{H_\gamma}{H_{\gamma\gamma}}\frac{1-\tau_k}{(1-\gamma_k)\tau_k} \tag{A.81}$$

As long as H_γ is log-concave, this expression is increasing in γ (as expected). If the marginal costs of avoidance are "too convex" in the avoidance rates, the curvature of avoidance costs rises fast enough to offset the shrinking taxable base. Log-concavity requires only that the proportional rate of increase of the marginal avoidance cost is diminishing, a standard regularity condition satisfied by common functional forms.

Under separability, $\frac{b_k}{G(r)}$ is independent/increasing/decreasing of b_k under CRS/IRS/DRS avoidance. Thus, using Lemma 3, this elasticity is independent of market power under CRS avoidance, increases with market power under IRS avoidance, and decreases with market power under DRS avoidance.

Enforcement: For the enforcement intensity, the reported taxable income response is

$$\frac{\partial Z_k}{\partial \beta_k} = \frac{1}{H_{\gamma\gamma}}\left[-H_{\gamma\beta} + (\tau_k - H_{b\gamma})\frac{\partial b_k}{\partial \beta_k}\right]B_k + (1-\gamma_k)\frac{\partial B_k}{\partial \beta_k} \tag{A.82}$$

Under a revenue tax with CRS avoidance technology,

$$\varepsilon^{\beta}_{Z_k} = -\frac{H_{\gamma\beta}}{H_{\gamma\gamma}}\frac{\beta_k}{1-\gamma_k} + \frac{\beta_k H_{b\beta}}{1-\omega_k}\frac{(1-\varepsilon_{x_k})}{\Delta}\left(1-\frac{\theta^c_k}{\varepsilon_{x_k}}\right) \tag{A.83}$$

Under a profit tax,

$$\varepsilon^{\beta}_{Z_k} = -\frac{H_{\gamma\beta}}{H_{\gamma\gamma}}\frac{\beta_k}{1-\gamma_k} \tag{A.84}$$

Log-concavity of H_γ implies that this expression is decreasing in γ_k. Separability eliminates any direct dependency on b (though it may indirectly depend on b_k via its impact on γ_k). In terms of magnitude, the relationship with this elasticity and market power are the same. This elasticity, however, is negative.

B.4 Optimal Levels (Proposition 2)

The government's Lagrangian is

$$\mathcal{L} = v(q) + \lambda \left[\sum_j \tau_j Z_j - N_j A(\beta_j, r_j) - G \right]$$

A common simplification will be to assume that the cost of enforcement is independent of the size of the firm: $A(\beta_j, r_j) = A(\beta_j, 1)$. One justification for this modeling decision is that there are often large fixed administrative costs to enforcement such that additional costs due to firm size are relatively small (Dharmapala et al., 2011). β_j then would be similar to an audit rate, which will scale up enforcement costs by the total potential audits (number of firms) in the industry.

Tax: Differentiating this expression with respect to the tax rate of an industry k gives

$$\alpha \left[-\frac{\partial q_k}{\partial \tau_k} x_k + \frac{\partial \Pi_k}{\partial \tau_k} \right] + \lambda \left[Z_k + \tau_k \frac{\partial Z_k}{\partial \tau_k} - A_b \frac{dB_k}{d\tau_k} \right] = 0$$

where the first bracket represents the welfare effect, and the second bracket represents the revenue effect. Dividing through by reported income gives

$$\frac{\alpha}{\lambda} \frac{1}{Z_k} \left[-\frac{\partial q_k}{\partial \tau_k} x_k + \frac{\partial \Pi_k}{\partial \tau_k} \right] + 1 - \frac{\tau_k}{1 - \tau_k} \left[\varepsilon_{Z_k} - \frac{A_b B_k}{\tau_k Z_k} \varepsilon_{B_k} \right] = 0$$

where the elasticities of reported and true taxable income are defined with respect to the retention rate $1 - \tau_k$. The ratio $\frac{A_b B_k}{\tau_k Z_k}$ can be interpreted as the marginal fraction of tax revenue used for enforcement costs. This is equal to zero when enforcement costs are fixed per firm. Using our work from previous section, we define

$$1 + EB_k = \frac{1}{Z_k} \left[\frac{\partial q_k}{\partial \tau_k} x_k - \frac{\partial \Pi_k}{\partial \tau_k} \right]$$

and thus, we can write the optimal tax rate in the simple form

$$\frac{\tau_k}{1 - \tau_k} = \frac{1 - \frac{\alpha}{\lambda} [1 + EB_k]}{\varepsilon_{Z_k} - \frac{A_b B_k}{\tau_k Z_k} \varepsilon_{B_k}}$$

From this point forward, we default to size-independent enforcement costs for cleanliness, but the general version will have this term subtracted off the reported income elasticity. The previous expression can be decomposed into the standard Ramsey rule and the market power component:

$$\frac{\tau_k}{1 - \tau_k} = \frac{1 - \frac{\alpha}{\lambda}}{\varepsilon_{Z_k}} - \frac{\frac{\alpha}{\lambda} EB_k}{\varepsilon_{Z_k}} \qquad \text{(A.85)}$$

This formulation highlights two often opposing consequences of monopolization. First, and perhaps more obvious, is that market power compounds pre-existing distortion, leading to additional excess burden. This unambiguously drives the optimal tax rate down. At the same time, market power drives firms up the demand curves toward more inelastic regions. This drives the optimal tax rate up. Thus the question whether market power increases or decreases the optimal tax rate hinges on the type of comparison. For two markets with the same taxable income elasticity, the less competitive market will have a lower optimal tax rate. For the same market (or a structurally identical market that only differs by market power), increasing market power has an ambiguous impact on the optimal tax rate.

Enforcement: Differentiating the Lagrangian with respect to enforcement gives

$$\alpha\left[-\frac{\partial q_k}{\partial \beta_k}x_k+\frac{\partial \Pi_k}{\partial \beta_k}\right]+\lambda\left[\tau_k\frac{\partial Z_k}{\partial \beta_k}-N_kA_\beta-A_b\frac{\partial B_k}{\partial \beta_k}\right]=0$$

which we can similarly convert to

$$-\frac{\alpha}{\lambda}\left[N_kH_\beta+Z_kEB_k^\beta\right]+\frac{\tau_k}{\beta_k}Z_k\varepsilon_{Z_k}^{\beta_k}-N_kA_\beta-\frac{A_b}{\beta_k}B_k\varepsilon_{B_k}=0$$

giving us

$$\beta_k=\frac{\tau_kz_k\varepsilon_{Z_k}^{\beta_k}-A_bb_k\varepsilon_{B_k}}{\left[A_\beta+\frac{\alpha}{\lambda}\left(H_\beta+z_kEB_k^\beta\right)\right]}$$

With fixed enforcement costs per-firm, this can be rearranged to a similar expression to that in Keen and Slemrod (2017):

$$\varepsilon_{Z_k}^{\beta_k}=\frac{\beta_k\left[A_\beta+\frac{\alpha}{\lambda}\left(H_\beta+z_kEB_k^\beta\right)\right]}{\tau_kz_K} \tag{A.86}$$

As described in their paper, this equates the enforcement elasticity with the marginal cost to revenue ratio. The RHS numerator represents the linear approximation of administrative and compliance costs with an additional excess burden term due to market power, while the denominator represents the tax base. Note that as the excess burden term increases, the enforcement rate must decrease. The excess burden term, however, does not fully capture the effects of monopolization. Also important is the reported income per firm: $\frac{Z_k}{N_k}$. The reason is due to economies of scale on both the administrative and compliance side.

Finally, we can insert the expression of the optimal tax rate into the above expressions to get the full joint optimum. The optimal enforcement intensity is

$$\beta_k\left[A_\beta+\frac{\alpha}{\lambda}\left(H_\beta+z_kEB_k^{\beta_k}\right)\right]=\frac{\left[1-\frac{\alpha}{\lambda}\left(1+EB_k\right)\right]}{\varepsilon_{Z_k}+1-\frac{\alpha}{\lambda}\left(1+EB_k\right)}\varepsilon_{Z_k}^{\beta_k}z_k \tag{A.87}$$

The bracket on the LHS is the marginal social cost of enforcement (per firm). The numerator on the RHS is the social-value weighted marginal tax revenue, while the denominator is net marginal cost of raising a dollar via taxes. Thus the optimal enforcement should completely internalize fiscal externalities and market power-related distortions. The optimal tax expression remains the same, though ε_{Z_k} and EB_k are evaluated at the above expression.

B.5 Optimal Profit Tax (Corollary 2.1)

Under a profit tax ($\mu = 1$), we have that $EB_k = EB_K^\beta = 0$. Thus, we can simplify our two prior expressions down to

$$\frac{\tau_k}{1-\tau_k} = \frac{1-\frac{\alpha}{\lambda}}{\varepsilon_{Z_k}}, \qquad \beta_k = \frac{\tau_k z_k \varepsilon_{Z_k}^{\beta_k}}{A_\beta + \frac{\alpha}{\lambda} H_\beta}$$

The statements on the relationship between avoidance technology, market power, and the optimal tax rates directly follow Lemma 3. Under a profit tax, the only reason to have differentiated tax rates is if there are differences in avoidance behavior. Under constant scale avoidance, we have that $\varepsilon_{Z_k} = \varepsilon_{\gamma_k}$ is independent of the market power within a market. Thus, the (conditional on enforcement) tax rate does not depend on market concentration. For increasing returns to scale avoidance technology, the behavioral response is larger for larger firms, and thus the tax rate is driven down as market power increases.

Optimal enforcement may still differ under constant scale avoidance due to non-proportional enforcement cost. In particular, the social marginal cost of enforcement per dollar of firm tax base is

$$\frac{A_\beta + \frac{\alpha}{\lambda} H_\beta}{b_k} \tag{A.88}$$

Under constant scale avoidance the second term, compliance costs, is independent of firm size. The first time, government enforcement costs, decrease with firm size. Increasing scale avoidance technology pushes the second term in the same direction, while decreasing scale technology pushes it in the opposite direction.

B.6 Tradeoff (Proposition 3)

The condition for a revenue neutral change in the tax rate given a change in the enforcement is

$$\left[\tau_k \frac{\partial Z_k}{\partial \tau_k} + Z_j - A_b \frac{\partial B_k}{\partial \tau_k}\right] \frac{d\tau_k}{d\beta_k}\bigg|_G + \tau_k \frac{\partial Z_k}{\partial \beta_k} - N_k A_\beta - A_b \frac{\partial B_k}{\partial \beta_k} = 0 \tag{A.89}$$

We can rearrange to get the required change in the tax as

$$\frac{d\tau_k}{d\beta_k}\bigg|_G = \frac{\frac{N_k}{Z_k} A_\beta - \frac{\tau_k}{\beta_k}\left(\varepsilon_{Z_k}^\beta - \frac{A_b B_k}{\tau_k Z_k} \varepsilon_{B_k}^\beta\right)}{1 - \frac{\tau_k}{1-\tau_k}\left(\varepsilon_{Z_k} - \frac{A_b B_k}{\tau_k Z_k} \varepsilon_{B_k}\right)} \tag{A.90}$$

The change in welfare $dW = \frac{\partial W}{\partial \tau_k} \frac{d\tau_k}{d\beta_k}\Big|_G + \frac{\partial W}{\partial \beta_k}$ is given by

$$dW = \left[-\frac{\partial q_k}{\partial \tau_k} + \frac{\partial \Pi_k}{\partial \tau_k} \right] \frac{d\tau_k}{d\beta_k}\Big|_G + \left[-\frac{\partial q_k}{\partial \beta_k} + \frac{\partial \Pi_k}{\partial \beta_k} \right] \tag{A.91}$$

From before, we have

$$\frac{dW}{d\tau_k} = -Z_k - \rho_k z_k q_k [\omega_k + \theta_k^c (1 - \omega_k)], \qquad \frac{dW}{d\beta_k} = -N_k H_\beta - \rho_k^\beta x_k q_k [\omega_k + \theta_k^c (1 - \omega_k)] \tag{A.92}$$

At the optimum $dW = 0$, giving the following expression governs the optimal tradeoff between taxation and enforcement in industry k:

$$\frac{\frac{\tau_k}{\beta_k} \left(\varepsilon_{Z_k}^\beta - \frac{A_b B_k}{\tau_k Z_k} \varepsilon_{B_k}^\beta - \frac{\beta_k}{\tau_k z_k} A_\beta \right)}{1 - \frac{\tau_k}{1-\tau_k} \left(\varepsilon_{Z_k} - \frac{A_b B_k}{\tau_k Z_k} \varepsilon_{B_k} \right)} = \frac{N_k H_\beta + \frac{dq_k}{d\beta_k} x_k [\omega_k + \theta_k^c (1 - \omega_k)]}{Z_k + \frac{dq_k}{d\tau_k} x_k [\omega_k + \theta_k^c (1 - \omega_k)]} \tag{A.93}$$

If, evaluated at a potentially non-optimal (τ_k, β_k), the LHS is greater than the RHS, then the government should raise additional revenue with stronger enforcement rather than a higher tax rate. The prescription is of course reversed if the LHS is smaller than the RHS. Under a profit tax, this simplifies to

$$\frac{\varepsilon_{Z_k}^\beta - \frac{\beta_k}{\tau_k z_k} A_\beta}{1 - \frac{\tau_k}{1-\tau_k} \varepsilon_{Z_k}} = \frac{\beta_k H_\beta}{\tau_k z_k} \tag{A.94}$$

which matches the corresponding expression for the consumer problem found in Keen and Slemrod (2017). These two expressions can demonstrate how market power changes things on the firm side of the problem. For similar elasticities of taxable income and enforcement, the relative advantage of enforcement increases as the average firm compliance cost to tax base $\frac{H_\beta}{z_k}$ decreases. Lemma [] gives the direction of the relationship of this term and market power.

When the government cannot use a pure profit tax, then output distortions enter the calculus. In addition to the above conclusions, this also factors in the relative size of the pass-through rate with respect to the enforcement rate and with respect to the tax rate. Under CRS avoidance, we have that the ratio of welfare effects is

$$\begin{aligned} \frac{\frac{dW}{d\tau_k}}{\frac{dW}{d\beta_k}} &= \frac{N_k H_\beta + x_k q_k \frac{H_{b\beta}}{(1-\omega_k)\Delta} \left(1 - \frac{\theta_k^c}{\varepsilon_{x_k}} \right) [\omega_k + \theta_k^c (1 - \omega_k)]}{Z_k + x_k q_k \frac{1-\gamma_k}{(1-\omega_k)\Delta} \left(1 - \frac{\theta_k^c}{\varepsilon_{x_k}} \right) [\omega_k + \theta_k^c (1 - \omega_k)]} \\ &= \frac{H_{rb\beta} \left(1 + \frac{1}{(1-\omega_k)\Delta} \left(1 - \frac{\theta_k^c}{\varepsilon_{x_k}} \right) [\omega_k + \theta_k^c (1 - \omega_k)] \right)}{1 + \frac{1}{(1-\omega_k)\Delta} \left(1 - \frac{\theta_k^c}{\varepsilon_{x_k}} \right) [\omega_k + \theta_k^c (1 - \omega_k)]} \\ &= H(1, \gamma, \beta) \end{aligned} \tag{A.95}$$

Thus, there is no size dependency in the welfare tradeoff. Any difference in the tax-to-enforcement ratio then stems entirely from the fiscal tradeoff. For two industries with observably similar income elasticities, so long as marginal administrative costs per firm do not scale proportionately or more to the size of the firm (A is fixed to concave in b), then the industry with larger firm sizes should have higher enforcement-to-tax ratios. This is due to the government taking advantage of administrative cost economies of scale. As we have shown, larger firm sizes are positively associated with market power under increased concentration, while dependent on the elasticity of demand under softer conduct.

B.7 Corollary 3.1

This largely follows from the previous proposition and the logic of Corollary 2.1, but we re-formalize it here again. The government's first order condition with respect to a change in the *profit tax* is

$$\alpha \frac{\partial \Pi_k}{\partial \tau_k} + \lambda \left[Z_k + \tau_k \frac{\partial Z_k}{\partial \tau_k} \right] = 0$$

which becomes

$$1 - \frac{\alpha}{\lambda} + \tau_k \frac{\partial (1-\gamma_k)}{\partial \tau_k} \frac{1}{1-\gamma_k} = 0$$

or

$$1 - \frac{\alpha}{\lambda} - \frac{\tau_k}{1-\gamma_k} \frac{b_k}{H_{\gamma\gamma}} = 0$$

Using the avoidance first order condition, this can finally be rewritten as

$$\frac{H_\gamma}{(1-\gamma_k) H_{\gamma\gamma}} = 1 - \frac{\alpha}{\lambda}$$

The LHS increases as γ increases so long as H is not too convex (log-concave is a sufficient condition), while the RHS is clearly constant. Thus, if γ is higher for a (non-tax) reason, it must be the case that the tax rate must decrease in order to bring γ back down and balance the equation. The relationship between the tax rate and market power follows from Lemma [].

The government's first order condition with respect to a change in the enforcement rate while administering a profit tax is

$$\alpha \frac{\partial \Pi_k}{\partial \beta_k} + \lambda \left[\tau_k \frac{\partial Z_k}{\partial \beta_k} - N_k A_\beta \right] = 0 \tag{A.96}$$

which becomes

$$-\frac{\alpha}{\lambda} N_k H_\beta + \tau_k \frac{H_{\gamma\beta}}{H_{\gamma\gamma}} B_k - N_k A_\beta = 0 \tag{A.97}$$

or simplified to

$$\tau_k \left[\frac{H_{\gamma\beta}}{H_{\gamma\gamma}} - \frac{\alpha}{\lambda} \frac{H_\beta}{H_\gamma} \right] = \frac{A_\beta}{b_k} \tag{A.98}$$

The term in the brackets is proportional to γ. Similar to before, if γ increases it must be the case that β_k must increase in order to bring γ down (or directly since the bracketed term is proportional to the inverse of β). Again using our previous expression on how γ^* changes with market, we can make the same conclusion on the relationship between market power and the enforcement rate. The relationships for the enforcement-tax ratio follows.

B.8 Uniform Taxation (Proposition 4)

Under uniform taxation, the Lagrangian is

$$\mathcal{L} = v(q) + \lambda \left[\sum_j \tau Z_j - N_j A(\beta_j, r_j) - G \right]$$

Differentiating this expression with respect to the tax rate, we get

$$\sum_j \alpha \left[-\frac{\partial q_j}{\partial \tau} x_j + \frac{\partial \Pi_j}{\partial \tau} \right] + \lambda \left[Z_j + \tau \frac{\partial Z_j}{\partial \tau} - N_j A_b \frac{\partial R_j}{\partial \tau} \right] = 0$$

which we can restate as

$$\sum_j \frac{\alpha}{\lambda} \frac{Z_j}{Z_j} \left[-\frac{\partial q_j}{\partial \tau} x_j + \frac{\partial \Pi_j}{\partial \tau} \right] + Z_j - \frac{\tau}{1-\tau} \left[Z_j \varepsilon_{Z_j} - \frac{A_b R_j}{\tau} \varepsilon_{R_j} \right] = 0$$

using the same definition of excess burden from before, we can rearrange this expression to get

$$\frac{\tau}{1-\tau} = \frac{\sum_j Z_j \left[1 - \frac{\alpha}{\lambda} [1 + EB_j] \right]}{\sum_j Z_j \left[\varepsilon_{Z_j} - \frac{A_b R_j}{\tau Z_j} \varepsilon_{R_j} \right]}$$

The uniform tax rate looks similar to the differentiated rate. The major difference is that the fiscal distortion and the market power burdens are weighted by reported income. The optimal tax rate falls if either the income elasticity or markets with high market-power related distortions are positively correlated with reported income. The enforcement side, however, has the same expression as before:

$$\beta_k = \frac{\tau z_k \left(\varepsilon^{\beta}_{Z_k} - \frac{A_b b_k}{\tau z_k} \varepsilon^{\beta}_{B_k} \right)}{A_\beta + \frac{\alpha}{\lambda} \left(H_\beta + z_k EB^{\beta}_K \right)} \tag{A.99}$$

The only change is to replace the differentiated tax rate with the uniform tax rate.

Under a uniform corporate tax rate, the ratio of optimal enforcement for markets m and n is given by

$$\frac{\varepsilon^{\beta_m}_{Z_m} z_m}{\varepsilon^{\beta_n}_{Z_n} z_n} = \frac{\beta_m \left[A_\beta(\beta_m) + \frac{\alpha}{\lambda} \left(H_\beta + z_m EB_m \right) \right]}{\beta_n \left[A_\beta(\beta_n) + \frac{\alpha}{\lambda} \left(H_\beta + z_n EB_n \right) \right]} \tag{A.100}$$

Under uniform profit tax, this becomes

$$\frac{\varepsilon_{Z_m}^{\beta_m} z_m}{\varepsilon_{Z_n}^{\beta_n} z_n} = \frac{\beta_m \left[A_\beta(\beta_m) + \frac{\alpha}{\lambda} H_\beta\right]}{\beta_n \left[A_\beta(\beta_n) + \frac{\alpha}{\lambda} H_\beta\right]} \tag{A.101}$$

Since under a profit tax

$$\varepsilon_{Z_k}^{\beta} = \frac{d(1-\gamma_k)}{d\beta_k} \frac{\beta_k}{(1-\gamma_k)} = \frac{H_{\gamma\beta}\beta_k}{(1-\gamma_k)H_{\gamma\gamma}} \tag{A.102}$$

This equation simplifies to

$$\frac{\frac{H_{\gamma\beta}(\gamma_m,\beta_m,1)}{H_{\gamma\gamma}(\gamma_m,\beta_m,1)}}{\frac{H_{\gamma\beta}(\gamma_n,\beta_n,1)}{H_{\gamma\gamma}(\gamma_n,\beta_n,1)}} = \frac{\left[A_\beta(\beta_m) + \frac{\alpha}{\lambda} H_\beta\right]}{\left[A_\beta(\beta_n) + \frac{\alpha}{\lambda} H_\beta\right]} \frac{b_n}{b_m} \tag{A.103}$$

The LHS is proportional to $\frac{\beta_n}{\beta_m}$. The RHS is independent of firm size if and only if both compliance costs and administrative costs are linear in firm size. If administrative costs exhibit economies of scale, and compliance costs weakly exhibit economies of scale, then the RHS > 1 if and only if $b_m > b_n$. This implies that $b_m > b_n \iff \frac{\beta_n}{\beta_m} > 1$. Industry m benefits more from enforcement related economies of scale, and therefore the government should enforce this industry more. In total, we just need that the sum function of administrative and compliance costs are concave in firm size.

In the differentiated tax case, the ratio condition is

$$\frac{\varepsilon_{Z_m}^{\beta_m} z_m}{\varepsilon_{Z_n}^{\beta_n} z_n} \frac{\tau_m}{\tau_n} = \frac{\beta_m \left[A_\beta(\beta_m) + \frac{\alpha}{\lambda} H_\beta\right]}{\beta_n \left[A_\beta(\beta_n) + \frac{\alpha}{\lambda} H_\beta\right]} \tag{A.104}$$

For CRS avoidance, differentiation of the tax rates has no impact on relative avoidance. For IRS avoidance, $\frac{\tau_m}{\tau_n} > 1$, and so this increases the magnitude of the LHS, which means that the RHS must increase. Thus, $\frac{\beta_m}{\beta_n}$ increases. For DRS, the reverse is true. Combining these statements with Proposition 2.1 gives the result.

C Simulation Appendix

C.1 Cost Specifications

For the simulations, we assume that firms have the following cost function:

$$C(y, b, \gamma, \beta) = cy + \beta\gamma^2 b^\rho \tag{A.105}$$

and the government's enforcement costs are

$$B(\beta, b) = \kappa\beta_j^\sigma b^\delta \tag{A.106}$$

Note that, unlike in the theory, we allow enforcement costs to be a function of firm size. We will show how this can impact the results. Under these specifications, the optimal γ_k for a given (τ_k, β_k) is

$$\gamma_k^*(\tau, \beta) = \frac{\tau_k}{2\beta_k} b_k^{1-\rho} \tag{A.107}$$

Then the derivatives of avoidance with respect to each of the two tax tools are

$$\frac{d\gamma_k}{d\tau_k} = \frac{b_k^{1-\rho}}{2\beta_k} + (1-\rho)\frac{\tau_k}{2\beta_k} b_k^{-\rho} \frac{db_k}{d\tau_k} \tag{A.108}$$

$$\frac{\partial\gamma_k}{\partial\beta_k} = -\frac{\tau_k}{2\beta_k^2} b_k^{1-\rho} + (1-\rho)\frac{\tau_k}{2\beta_k} b_k^{-\rho} \frac{db_k}{d\beta_k} \tag{A.109}$$

Under a pure profit tax, b_k is independent of both the tax and the enforcement rate, and the second term drops out of both expressions. The FOCs for the government's problem are

$$1 - \frac{\alpha}{\lambda} + \tau_k \frac{\partial(1-\gamma_k)}{\partial\tau_k} \frac{B_k}{Z_k} = 0 \tag{A.110}$$

$$N_k \left(\frac{\alpha}{\lambda}\gamma_k^2 b_k^\rho + \sigma\kappa\beta_k^{\sigma-1} b_k^\delta\right) = \tau_k \frac{\partial(1-\gamma_k)}{\partial\beta_k} B_k \tag{A.111}$$

The enforcement condition can be rewritten as

$$\frac{\alpha}{\lambda}\frac{\tau_k^2}{4\beta_k^2} b_k^{2-\rho} + \sigma\kappa\beta_k^{\sigma-1} b_k^\delta = \frac{\tau_k^2}{2\beta_k^2} b_k^{2-\rho} \tag{A.112}$$

and then rearranged as

$$\sigma\kappa\beta_k^{\sigma-1} b_k^\delta = \left(2 - \frac{\alpha}{\lambda}\right)\frac{\tau_k^2}{4\beta_k^2} b_k^{2-\rho} \tag{A.113}$$

Thus, the optimal level of enforcement conditional on a tax rate is

$$\beta_k = \left[\frac{\left(2 - \frac{\alpha}{\lambda}\right)\tau_k^2}{4\sigma\kappa}\right]^{\frac{1}{\sigma+1}} b_k^{\frac{2-\rho-\delta}{\sigma+1}} \tag{A.114}$$

Similarly, the tax condition can be rewritten as

$$\left(1 - \frac{\alpha}{\lambda}\right)\left(1 - \frac{\tau_k}{2\beta_k} b_k^{1-\rho}\right) - \frac{\tau_k}{2\beta_k} b_k^{1-\rho} = 0 \tag{A.115}$$

which can be rearranged to

$$1 - \frac{\alpha}{\lambda} = \left(2 - \frac{\alpha}{\lambda}\right)\frac{\tau_k}{2\beta_k} b_k^{1-\rho} \tag{A.116}$$

and finally the optimal level of taxation conditional on an enforcement rate is

$$\tau_k = 2\beta_k \frac{1 - \frac{\alpha}{\lambda}}{2 - \frac{\alpha}{\lambda}} b_k^{\rho-1} \tag{A.117}$$

Finally, we solve jointly to get the joint optimum for the tax and enforcement rates as

$$\tau_k^* = \left[\frac{\left(1-\frac{\alpha}{\lambda}\right)^2}{\sigma\kappa\left(2-\frac{\alpha}{\lambda}\right)}\right]^{\frac{1}{\sigma-1}} \frac{2\left(1-\frac{\alpha}{\lambda}\right)}{2-\frac{\alpha}{\lambda}} b_k^{\frac{\rho-\delta}{\sigma-1}+\rho-1} \tag{A.118}$$

$$\beta_k^* = \left[\frac{\left(1-\frac{\alpha}{\lambda}\right)^2}{\sigma\kappa\left(2-\frac{\alpha}{\lambda}\right)}\right]^{\frac{1}{\sigma-1}} b_k^{\frac{\rho-\delta}{\sigma-1}} \tag{A.119}$$

Thus, we find that with a differentiated system, enforcement provides no targeting value if avoidance costs and enforcement costs scale with firm size at the same rate ($\rho = \delta$). The tax may still provide differentiation benefits so long as avoidance costs do not scale proportionally with firm size ($\rho = 1$). If $\rho = \delta = 1$, then there is no reason to have any differentiation in the tax system. As firm size provides no additional information on avoidance behavior or relative scaling of avoidance versus enforcement costs.

Under a uniform tax, the (only) tax condition is

$$\sum_j \left[\left(1-\frac{\alpha}{\lambda}\right)\left(1-\frac{\tau}{2\beta_j}b_k^{1-\rho}\right) - \frac{\tau}{2\beta_j}b_j^{1-\rho}\right] B_j = 0 \tag{A.120}$$

which simplifies to

$$\sum_j \left[\left(1-\frac{\alpha}{\lambda}\right) - \left(2-\frac{\alpha}{\lambda}\right)\frac{\tau}{2\beta_j}b_j^{1-\rho}\right] B_j = 0 \tag{A.121}$$

Substituting the enforcement condition (where τ_k is replaced with just τ), and solving for the full optimum tax rate and (differentiated enforcement) gives

$$\tau = \mathcal{A}^{\frac{\sigma+1}{\sigma-1}} \left[\frac{\sum_j N_j b_j}{\sum_j N_j b_j^{1+\eta}}\right]^{\frac{\sigma+1}{\sigma-1}} \tag{A.122}$$

$$\beta_k = A\mathcal{A}^{\frac{2}{\sigma-1}} \left[\frac{\sum_j N_j b_j}{\sum_j N_j b_j^{1+\eta}}\right]^{\frac{2}{\sigma-1}} b_k^{\frac{2-\rho-\delta}{\sigma+1}} \tag{A.123}$$

where

$$\eta = \frac{\sigma(1-\rho)+\delta-1}{\sigma+1}, \quad A = \left[\frac{2-\frac{\alpha}{\lambda}}{4\sigma\kappa}\right]^{\frac{1}{\sigma+1}}, \quad \mathcal{A} = 2A\frac{1-\frac{\alpha}{\lambda}}{2-\frac{\alpha}{\lambda}} \tag{A.124}$$

We can see from these expressions that if $\eta = 0$, then the optimal tax rate is independent of the distribution of firms. This is also exactly the condition under which a uniform tax system is optimal. There is no benefit to differentiate taxation to target firms by size, and therefore, even if unconstrained, the government should choose a singular tax rate.

The enforcement expression also implies that

$$\frac{\beta_m}{\beta_n} = \left(\frac{b_m}{b_n}\right)^{\frac{2-\rho-\delta}{\sigma+1}} \tag{A.125}$$

C.2 Additional Figures

C.2.1 Profit Tax, DRS

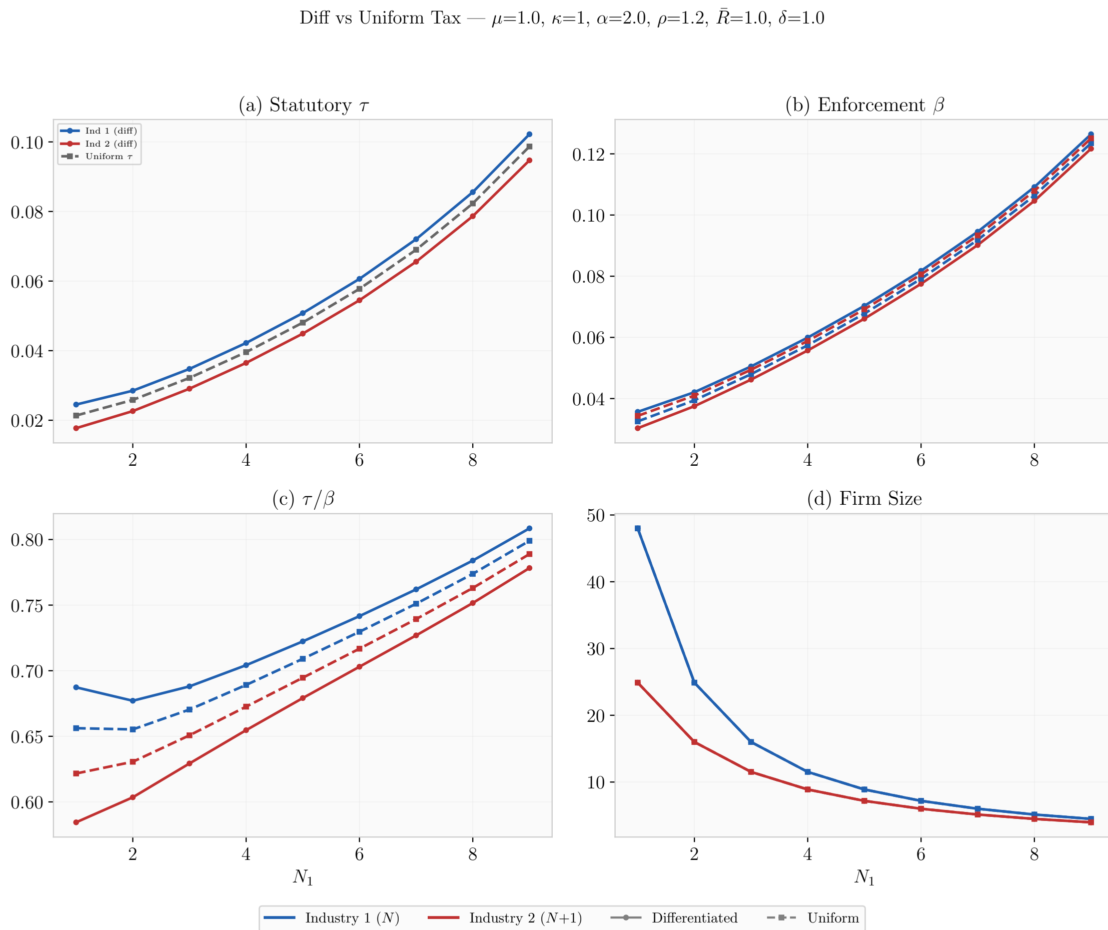

Figure 5: Optimal Policy by Number of Firms under Profit Taxation and DRS Avoidance

C.2.2 Revenue Tax, IRS

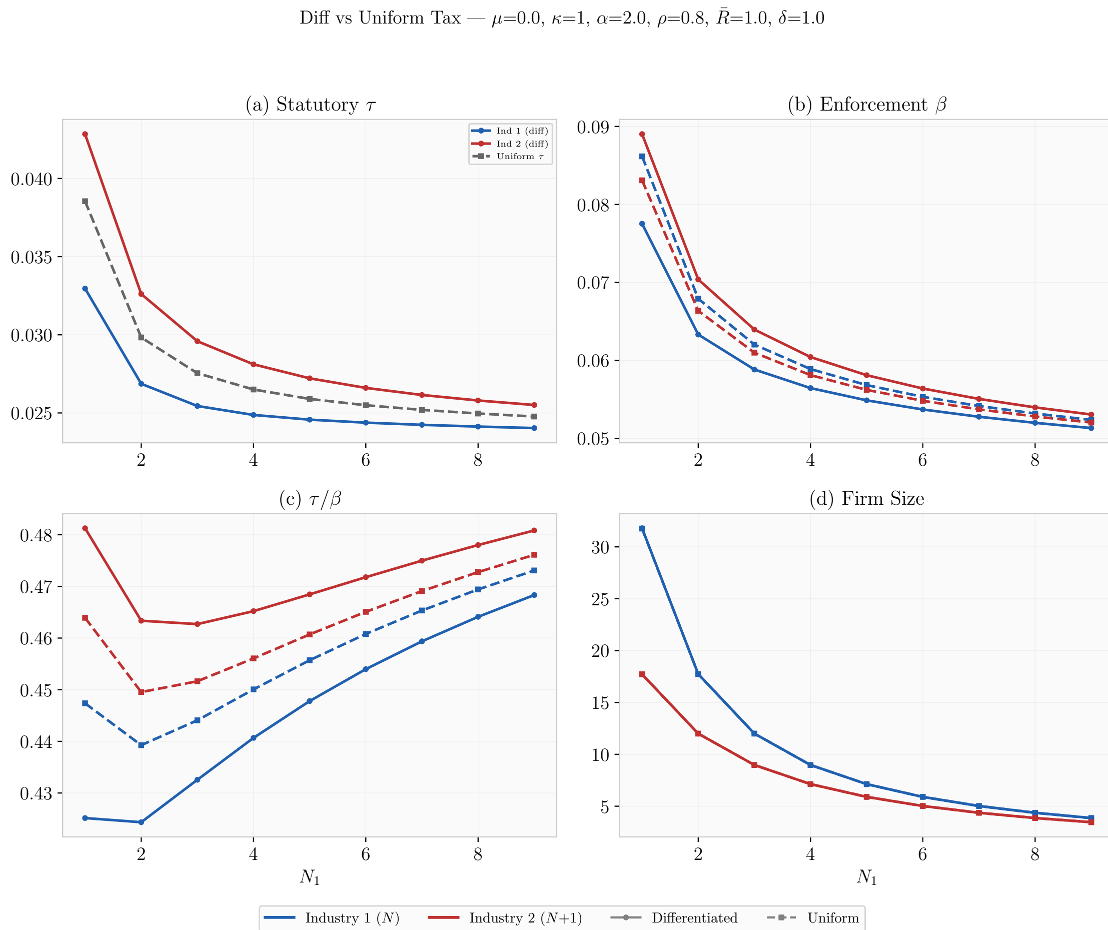

Figure 6: Optimal Policy by Number of Firms under Profit Taxation and IRS Avoidance

C.2.3 Profit Tax, IRS, Conduct Parameter

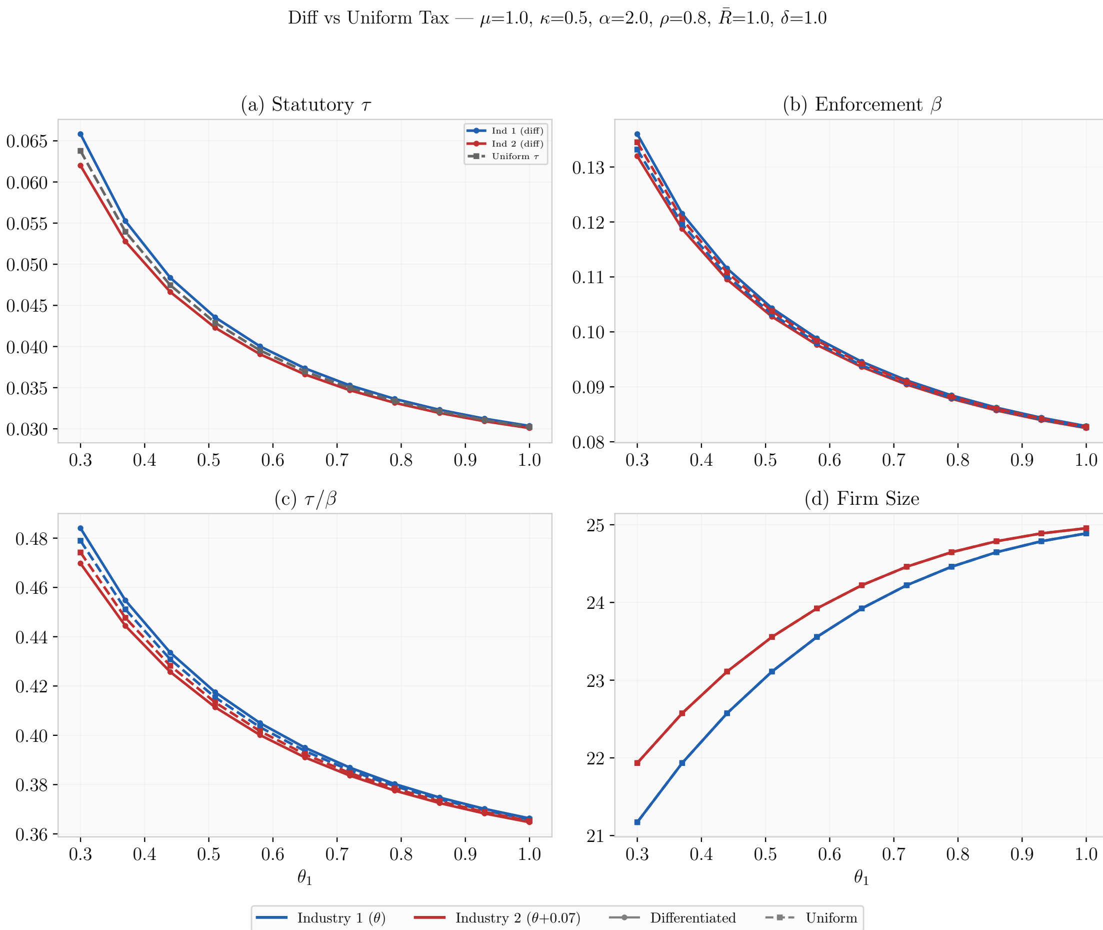

Figure 7: Optimal Policy by Conduct under Revenue Taxation and CRS Avoidance

C.2.4 Revenue Tax, CRS, Conduct Parameter

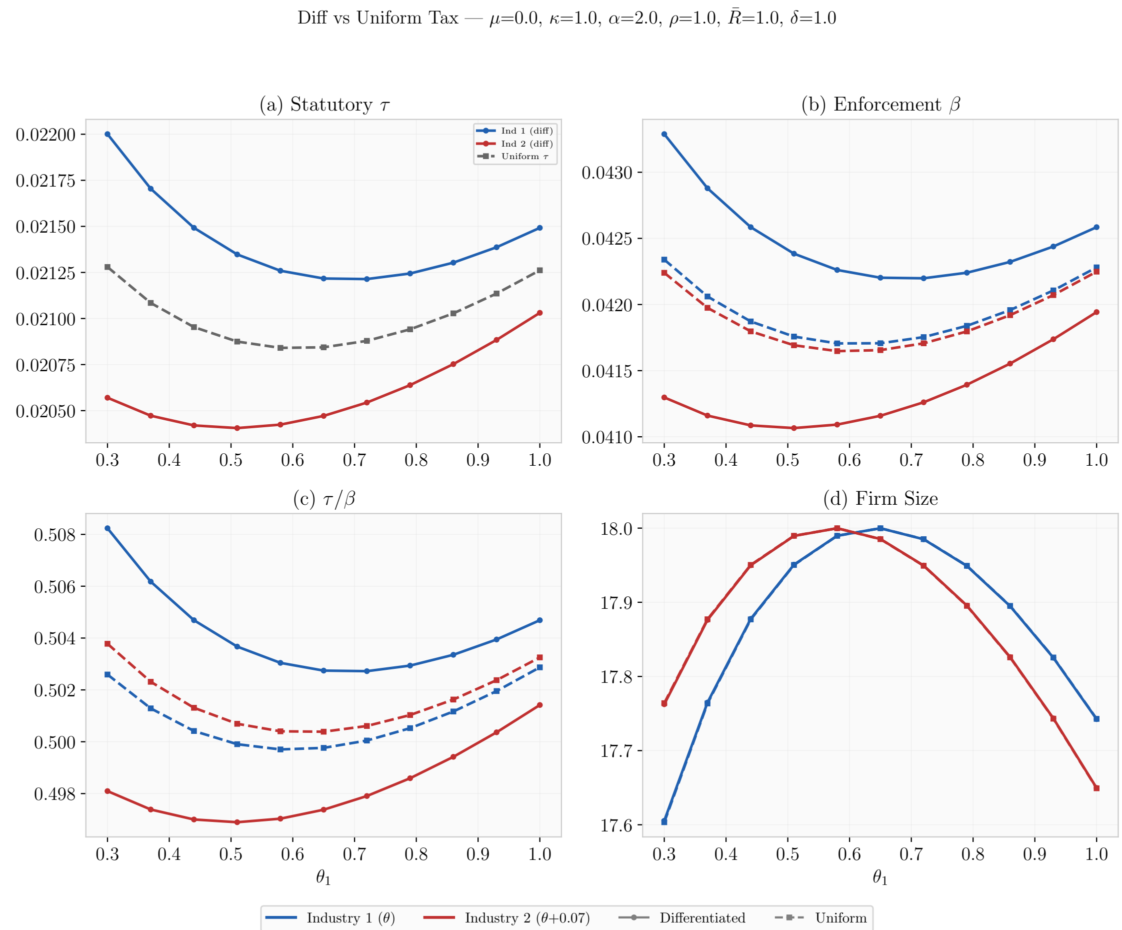

Figure 8: Optimal Policy by Conduct under Revenue Taxation and CRS Avoidance